

ANIME SCORE ANALYSIS

THE POWER OF DATA PREPARATION
THROUGH DATA STORYTELLING

Presented by

Group 1

PART 1

DATA STORYTELLING



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ACT 1

THE CONTEXT



1. BUSINESS CONTEXT



THE STRATEGIC GAP



Ambition

To decode the hidden formula behind high Audience Scores and build a predictive engine to validate those insights



Problem

The raw dataset is not ready for machine learning. It is structurally broken, filled with unstructured text, ghost entries, and misleading categories.

→ Using this data "as-is" will lead to incorrect business decisions.



2. DATASET PROFILE

DATASET PROFILE

Anime Dataset 2013

24,905
entries

24
features

anime_id	Name	English name	Other name	Score	Genres	Synopsis	Type	Episodes	Aired	Premiered	Status
1	Cowboy Bebop	Cowboy Bebop	カウボーイビバップ	8.75	Action, Award Winning, Sci-Fi	Crime is timeless. By the year 2071, humanity has expanded across the galaxy, filling the surface of	TV	26	Apr 3, 1998 to Apr 24, 1999	spring 1998	Finished Airing
Producers	Licensors	Studios	Source	Duration	Rating	Rank	Popularity	Favorites	Scored By	Members	Image URL
Bandai Visual	Funimation, Bandai Entertainment	Sunrise	Original	24 min per ep	R - 17+ (violence & profanity)	41	43	78525	914193	1771505	https://cdn.myanimelist.net/images/anime/4/19644.jpg

3. THE CONFLICT



THE CONFLICT

Raw Dataset Issues



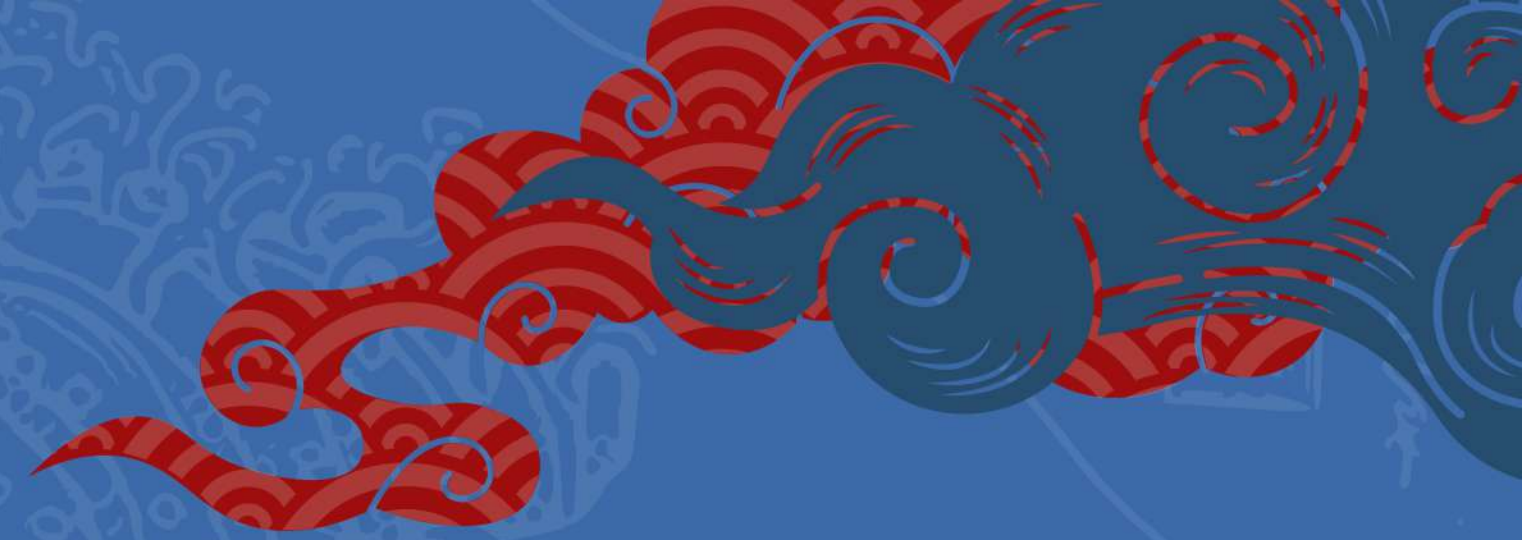
Critical Information Blind Spots



Structural Entropy



Signal Distortion



THE CONFLICT

Critical Information Blind Spots

→ A hole in our market understanding

Large portions of the dataset **pretends** to exist

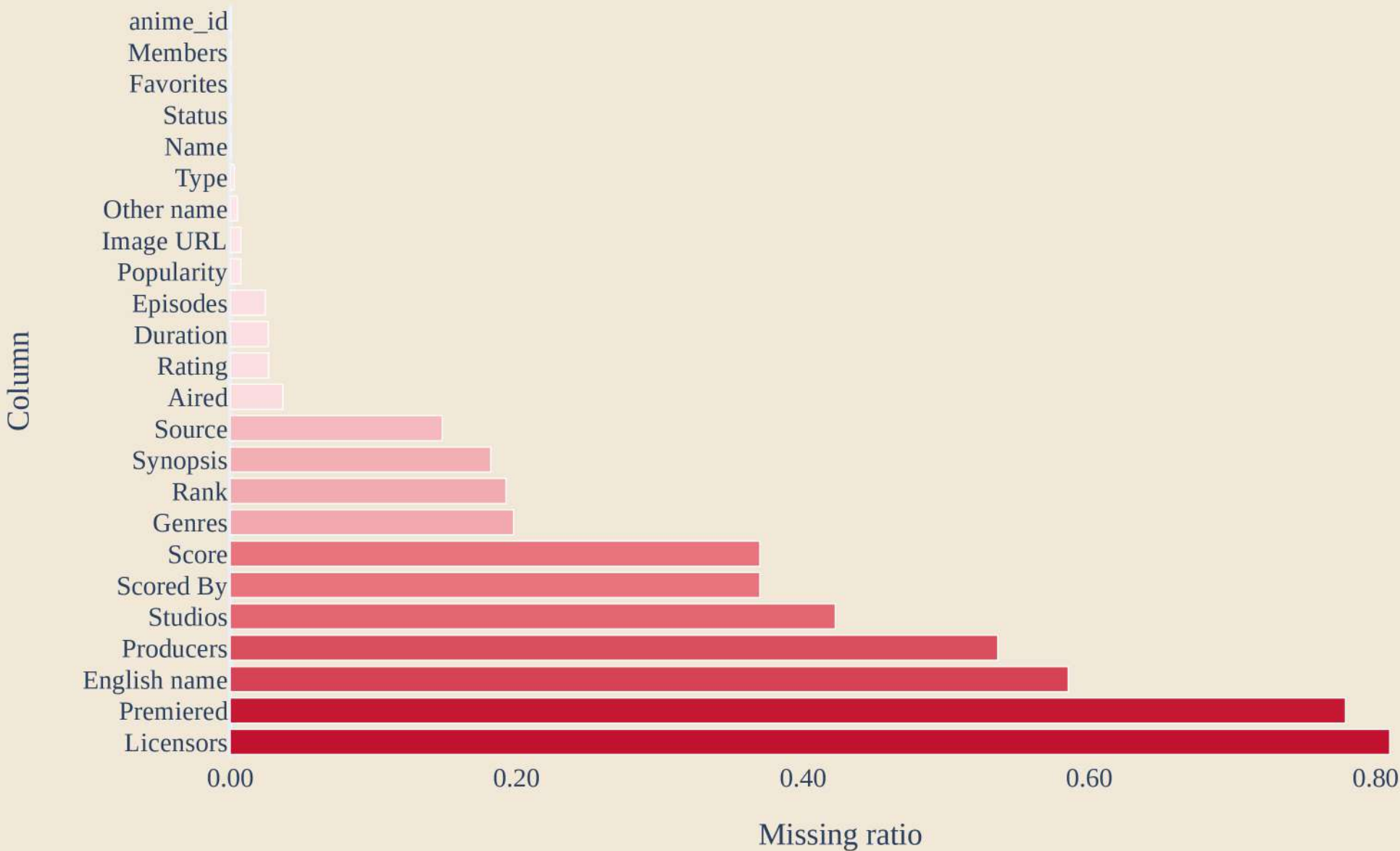
No description available for this anime.
Unknown Not available 0
<https://cdn.myanimelist.net/img/sp/icon/apple-...>

Score and Rank **contradict** each other

Rank = 190 Score = UNKNOWN

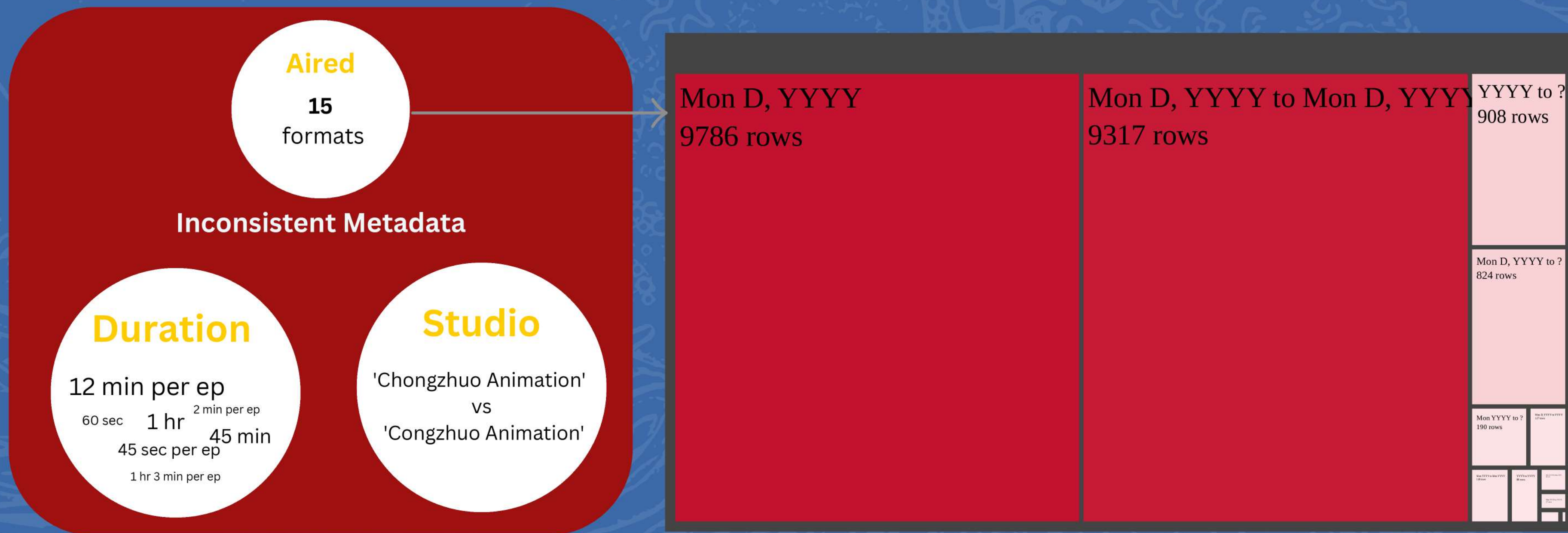
Columns with high missing data require cleaning

(Missing data ratio by column)



Inconsistent formats prevent direct comparison.

Without standardization, a machine learning model cannot interpret time or duration.



THE CONFLICT



Structural Entropy

Fragmented Metadata

- **Multi-label fields** explode into hundreds of rare or unique combinations.

Misclassified Metadata

Type, Episodes, and Duration contradict each other.

- Movies with multiple episodes
- TV entries with only a few minutes per episode
- Extreme episode counts

THE CONFLICT



Signal Distortion

→ Create false averages that mask the performance of real anime

Key engagement metrics (Members, Favorites, Episodes) are **heavily right-skewed**, making raw distributions statistically unstable.

Multi-entity fields contain **ghost titles with zero engagement and duplicate names** masked by inconsistent metadata.



4. CHAOS HANDLING

CHAOS HANDLING

Stage 1: Fixing Missingness

Expose true missing values, remove placeholders, restore critical fields.

Stage 2: Text and Structure Standardization

Normalize genres, studios, producers, and type labels.

Stage 3: Time Field Reconstruction

Parse Duration and Aired into consistent numerical formats.

Stage 4: Conflict Resolution

Correct misclassified Types, Episodes, and Durations.

Stage 5: Outlier and Distribution Repair

Identify extreme values and stabilize skewed metrics.

RAW DATA CREATES ILLUSIONS. ONLY PREPARED DATA REVEALS THE TRUTH.

書き初め

A single message that guides the entire story.





ACT 2

THE REVELATION

COMPARATIVE ANALYSIS

*From Chaos to Clarity: A Journey of
Discovery on Clean Data*



Target Variable



Market Factors



Creative & Production Factors



Release Strategy

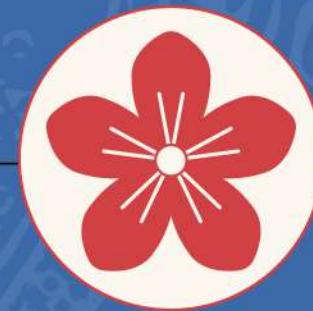
ANALYSIS STEPS



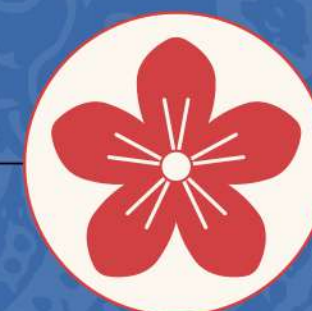
**Issues
Overview**



Solution



**True
Insight**



**Business
Insight**



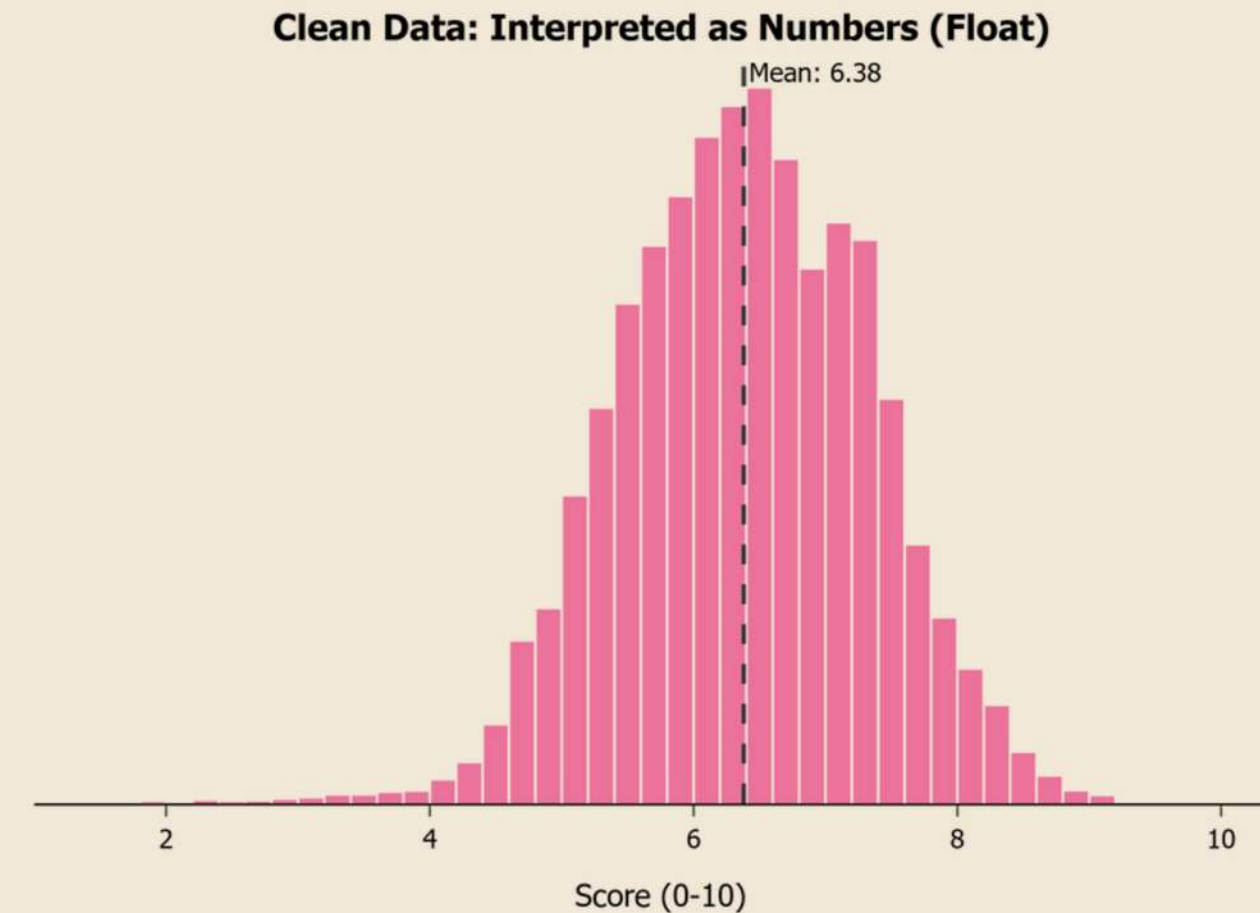
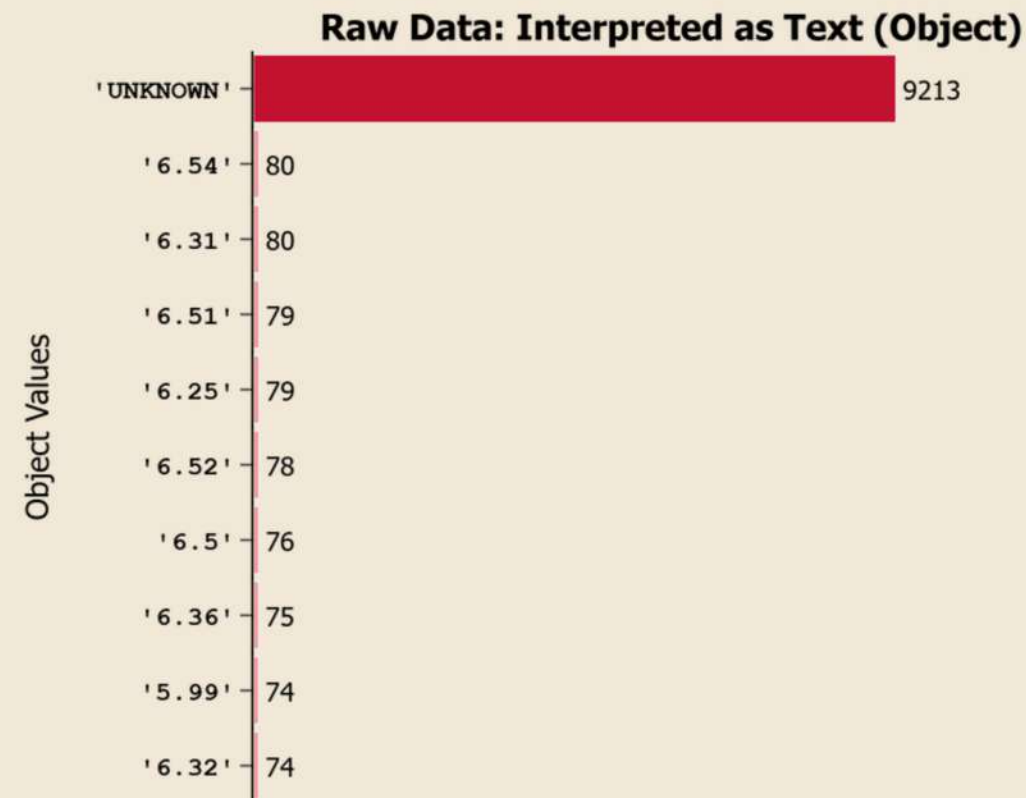
1. TARGET VARIABLE - SCORE



1.UNLOCKING A PREDICTABLE DISTRIBUTION ON CLEAN DATA

Converting 'Text Strings' into 'Calculable Metrics'

Left: Raw Score is treated as discrete categories (Text) | Right: Clean Score is a continuous variable (Numbers)



Insight

After coercion and filtering, a
Gaussian distribution emerges.
→ Measurable, predictable target



2. MARKET FACTORS

1

Type

Format types
(TV, Movie, etc.)

2

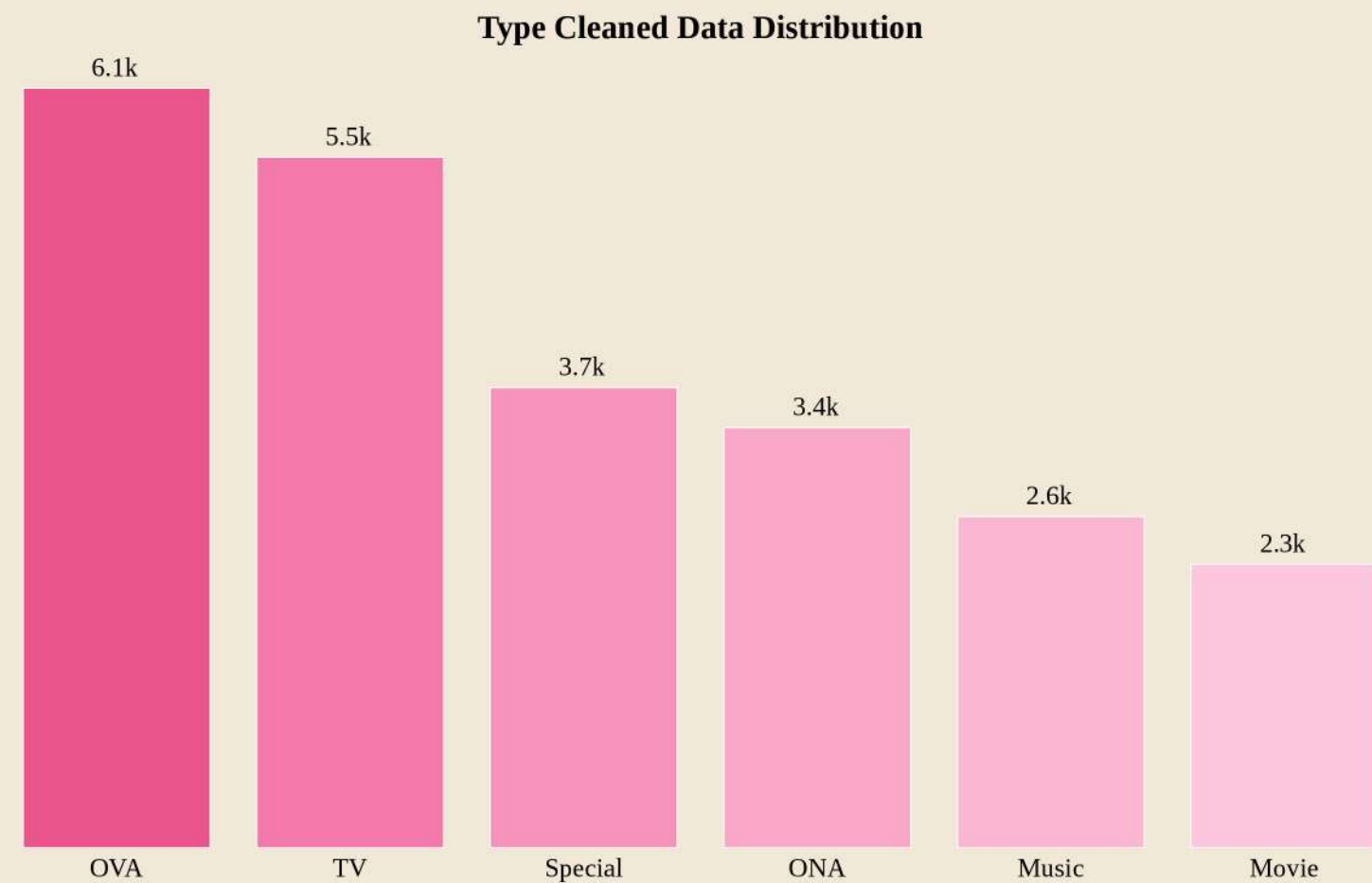
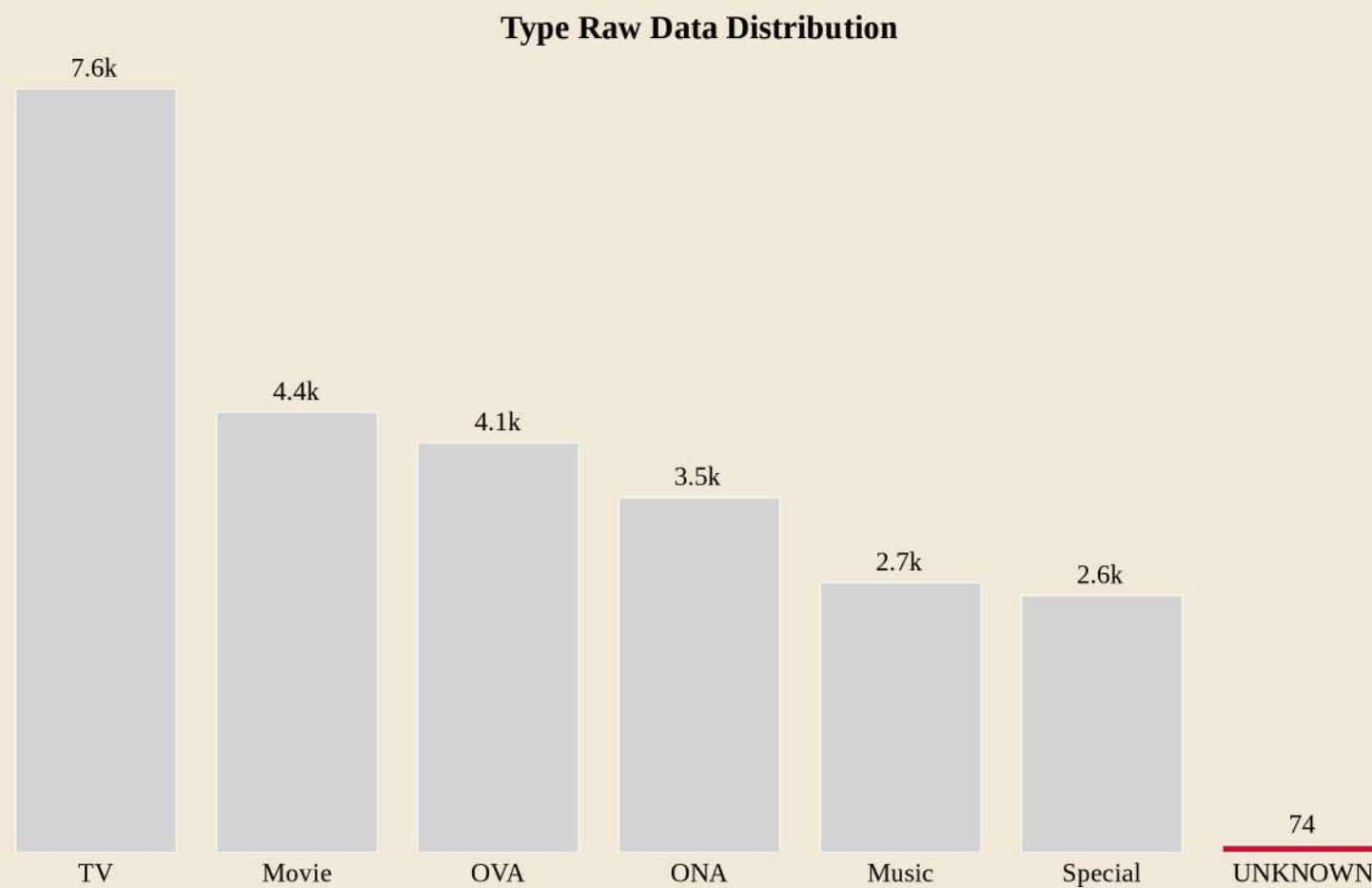
Source

Original source materials
(Manga, Novel, etc.)

2.1. TYPES - STANDARDIZATION REVEALS THE 'OVA' REALITY

Removing 'Unknown' Values Reveals True Type Distribution

(Top 10 Most Common Anime Types)

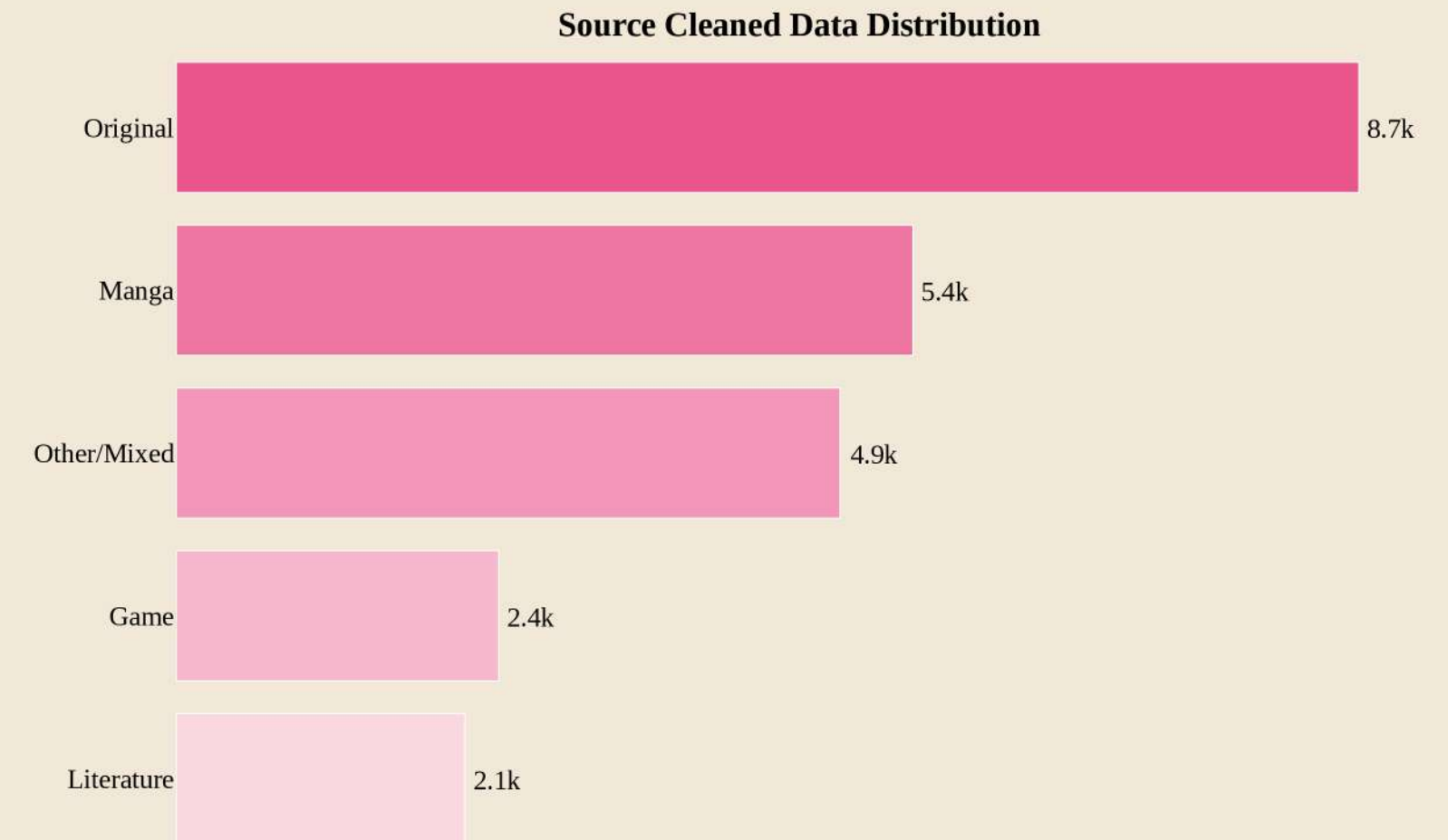
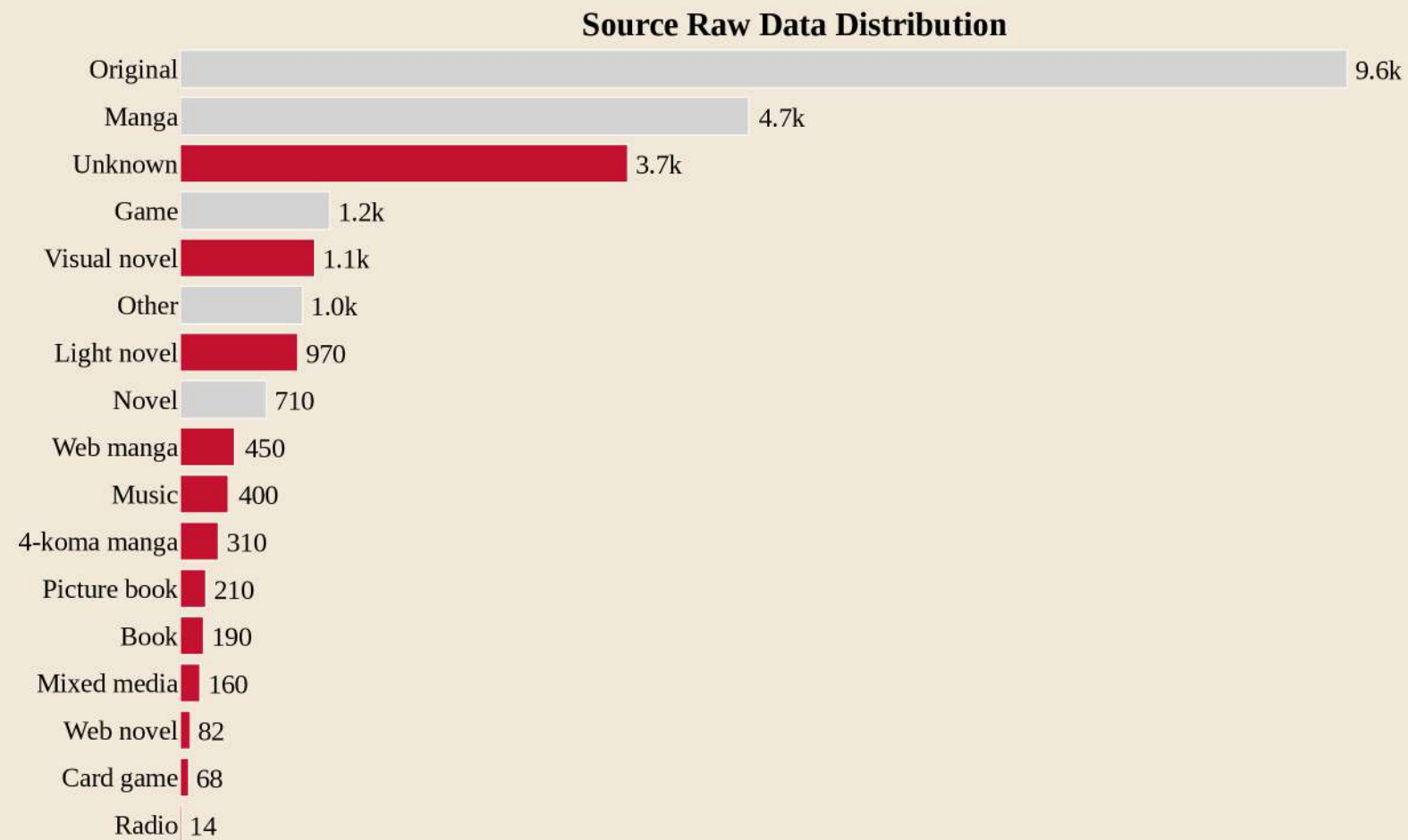


Insight

→ Data Preparation corrected our view of the market landscape that **OVA** is actually the most popular format.

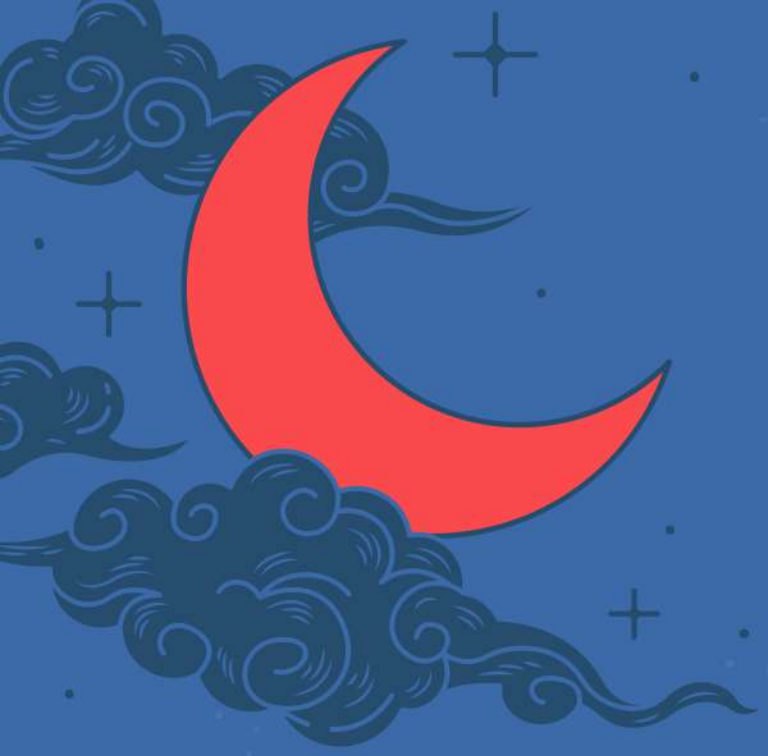
2.2.SOURCES - CLEAN DATA UNPACKED THE TRUE MARKET SHARE

Consolidating Fragmented Sources Reveals True Market Structure



Insight

Cleaning confirms **“Original”** as the **#1** source, correcting prior data dilution and delivering a reliable baseline for comparison.



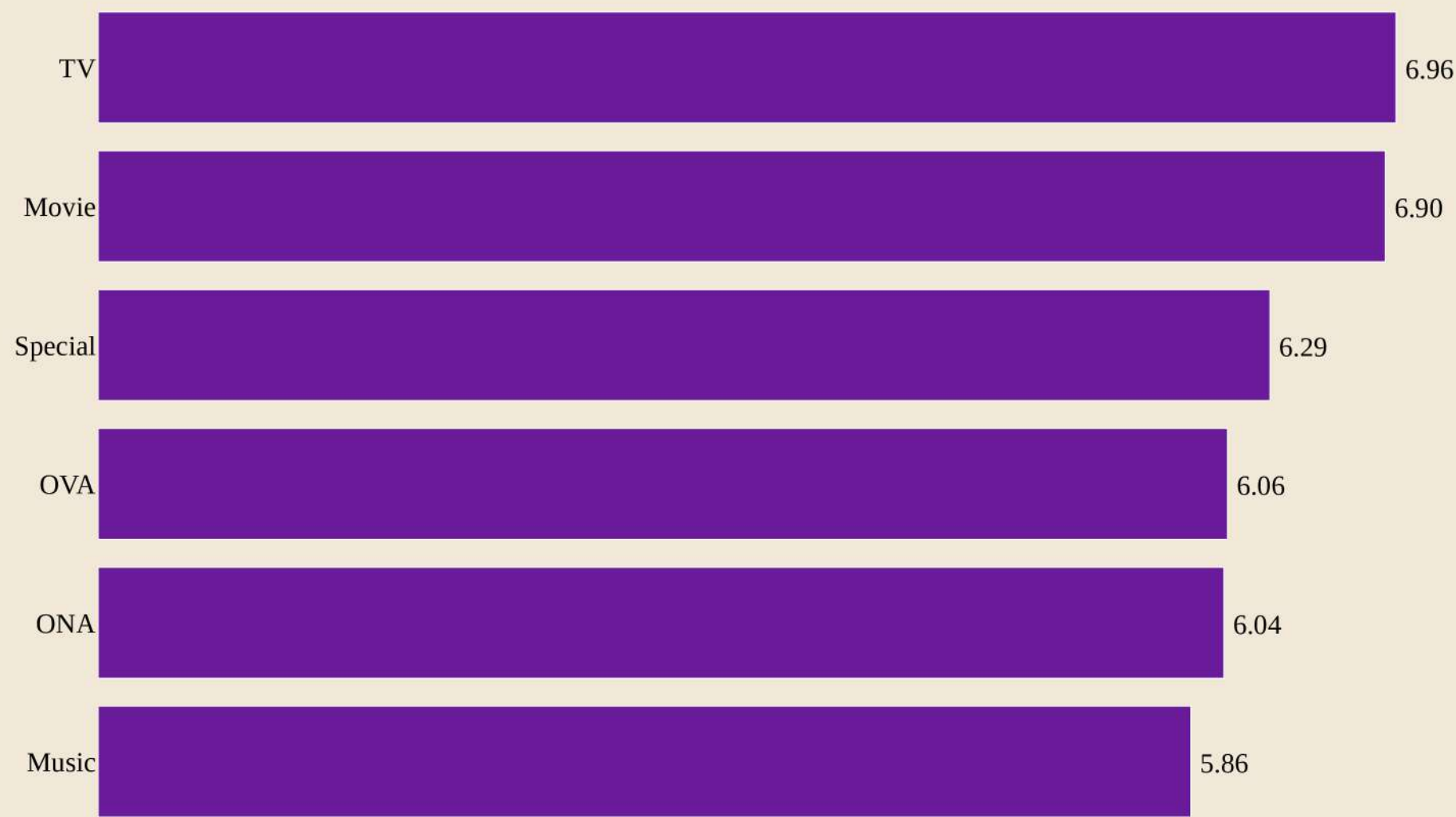
2.3. BUSINESS INSIGHT

2.3.1. VALIDATING FORMAT PERFORMANCE

Value of Clean Data: Identifying the “Quality Ceiling”

Movies and TV Series Lead in Average Audience Scores

(Average Score by Anime Type - All Formats Compared)



1. Data Confirmation: By filtering out misclassified short clips, the **cleaned dataset** confirms a distinct **'Quality Hierarchy'**. **TV Series and Movies statistically outperform** all other formats.

2. Structural Insight: The data proves that **OVA and ONAs** face a **structural score ceiling (<6.50)**, a trend that was **obscured** by **noise** in the **raw data**

2.3.1. VALIDATING FORMAT PERFORMANCE

Business Takeaways

- Concentrate resources on **TV & Movies** to break the **7.00 Score Barrier**.
- Treat **OVA**s/**ONA**s strictly as **fan engagement tools**, as they have a structural quality cap.

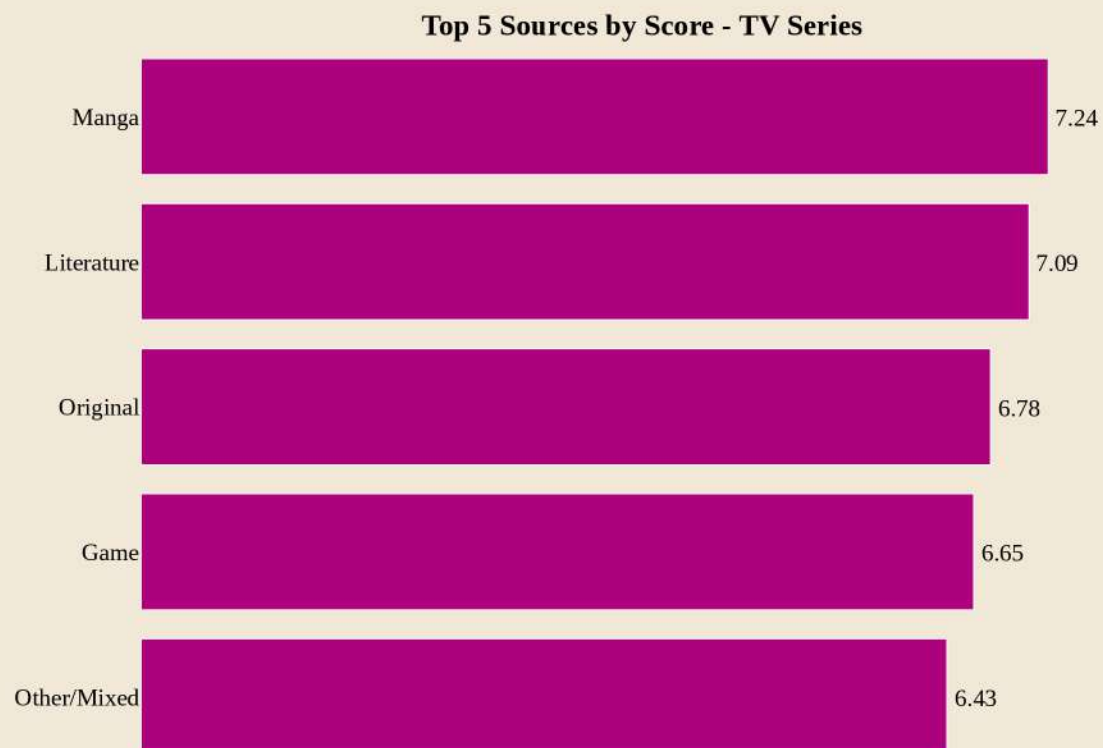


2.3.2. VALIDATING SOURCE MATERIAL IMPACT

Which source materials consistently deliver the highest quality perception (Score)?

Manga and Novel Adaptations Lead Quality Rankings Across Both Formats

(Top 5 Sources with Highest Average Scores - Minimum 30 Titles)



1. Aggregation Value: Consolidating **16 fragmented source types** into **5 macro-categories**
→ **Manga & Novels** provide a **statistically safer baseline** for success.

2. Risk Identification: The data explicitly highlights '**Game Adaptations**' as a **low-performance segment**, guiding the model to penalize this feature.

2.3.1. VALIDATING SOURCE MATERIAL IMPACT

Business Takeaways

- **Priority: Manga or Literature** maximizes score potential.
- **Strategy:** If **Original IP**, prioritize the **Movie** format over TV Series.
- **Caution:** Game adaptations face a historical quality cap; they require extra **investment** in **screenwriting**.





3. CREATIVE & PRODUCTION FACTORS

1

Genres

(Comedy, Fantasy, Romance, etc.)

2

Producers

Produce or fund the anime

3

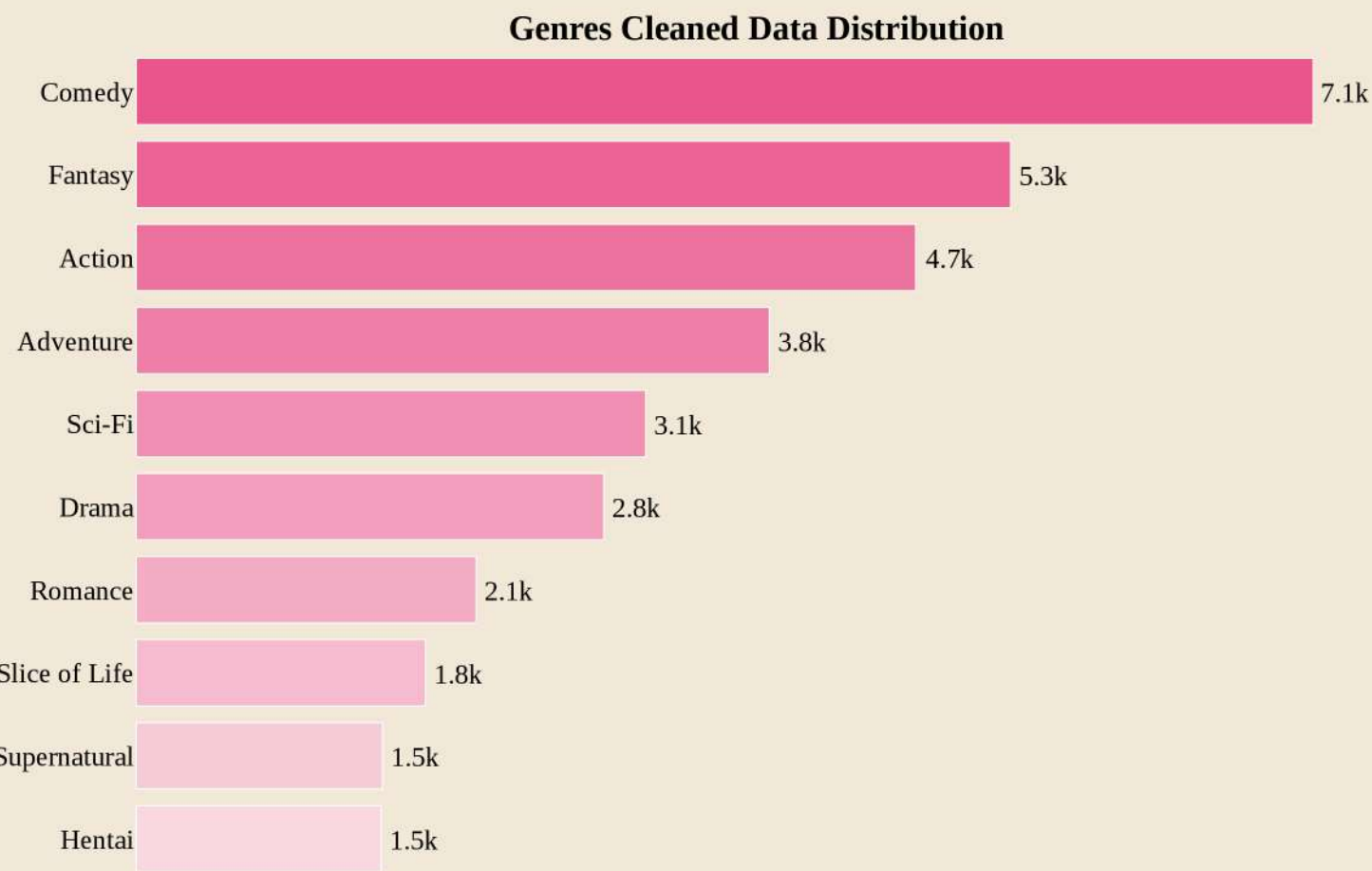
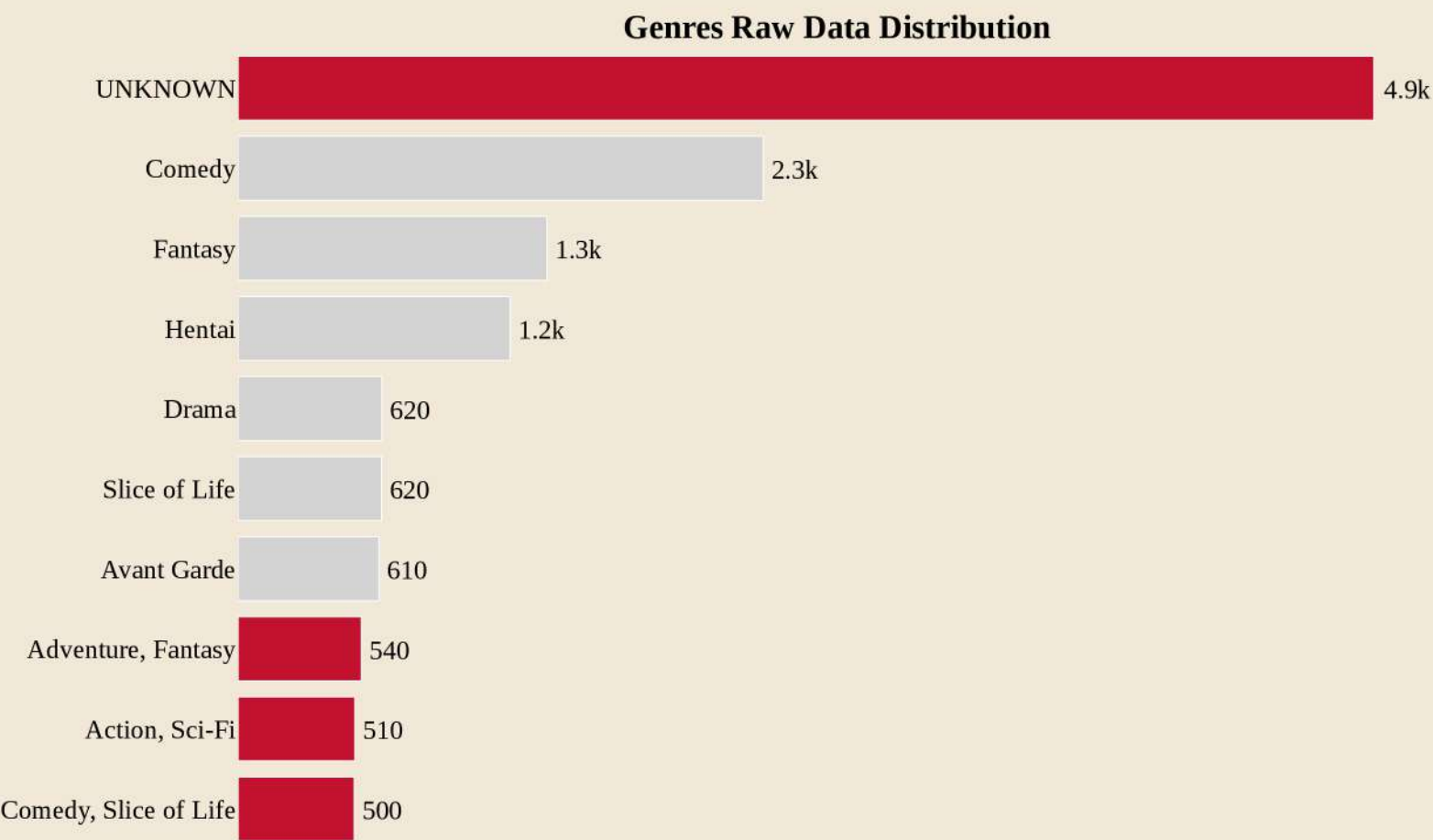
Studios

Directly responsible for visual content

3.1.GENRES - LIST EXPLOSION REVEALS 'COMEDY' DOMINANCE ON

Removing Aggregation & 'Unknown' Values Unlocks True Genres Distribution

(Top 10 Most Aired Genres By Number of Anime)



Insight

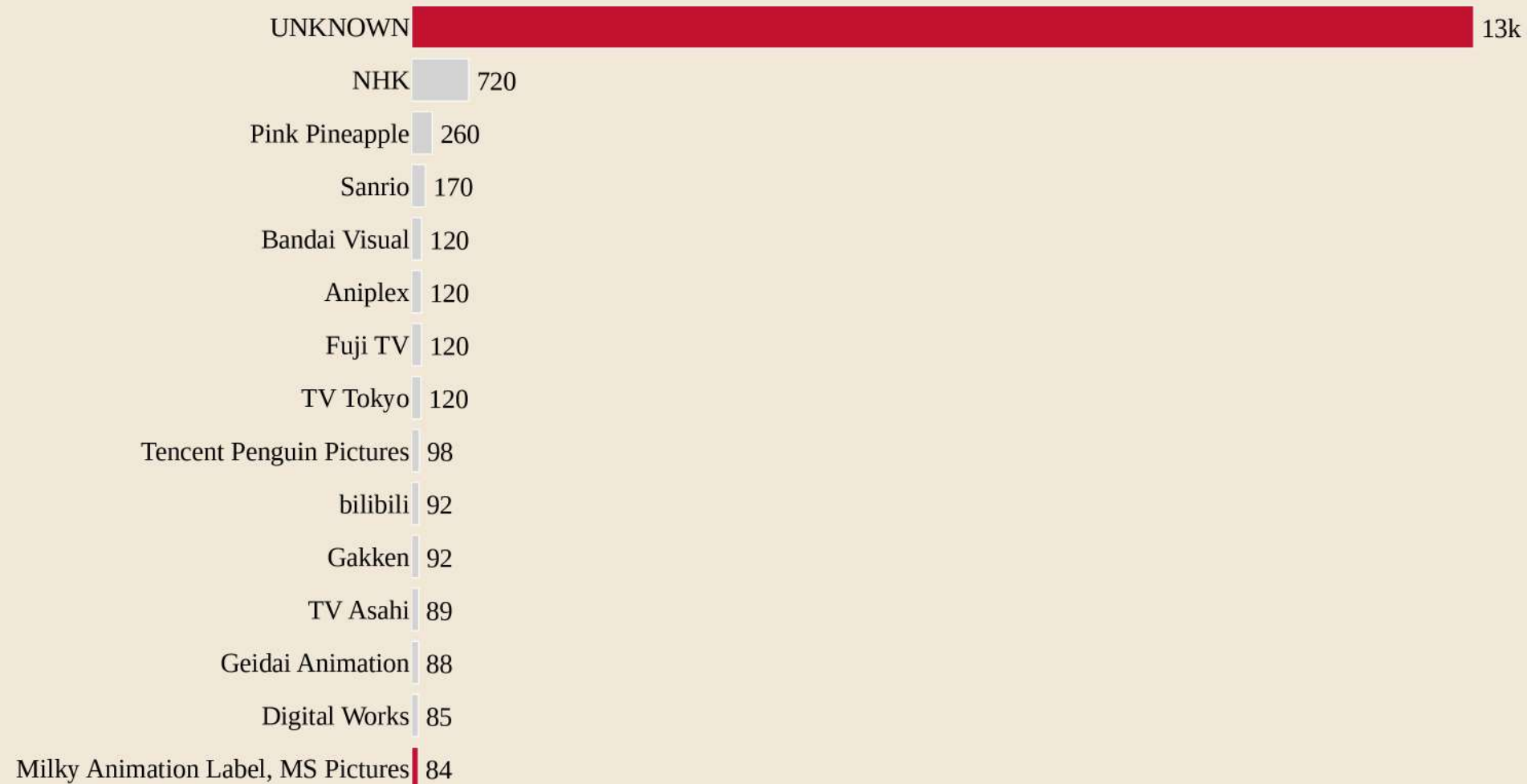
Multi-label binarization separates combined genre strings, removing false frequency inflation. The cleaned view reveals **'Comedy'** as the real top genre.

3.2. PRODUCERS - THE TRUE TOP STUDIO

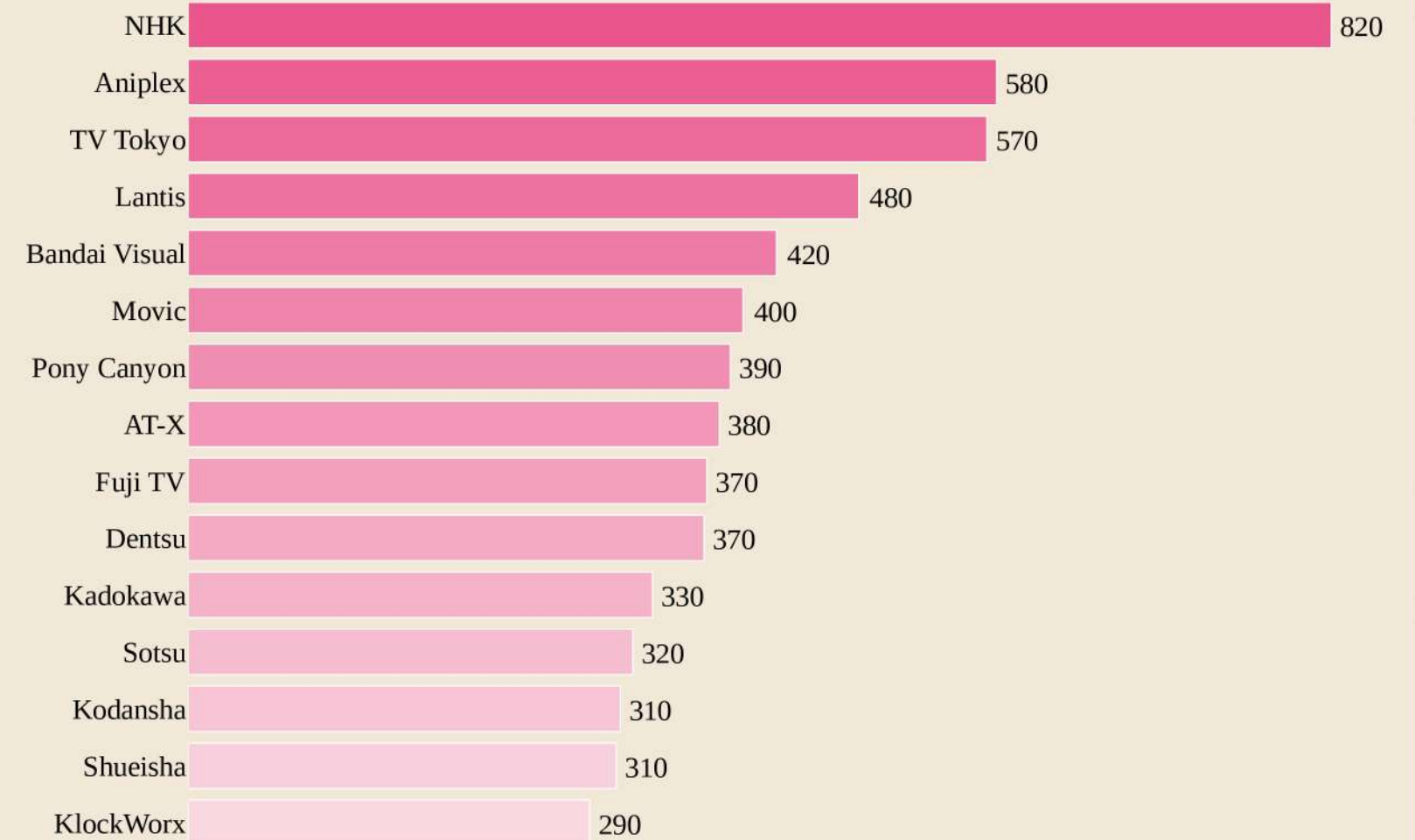
Parsing Producers Data Reveals True Market Leaders

(Top 15 Most Active Producers)

Producers Raw Data Distribution



Producers Cleaned Data Distribution

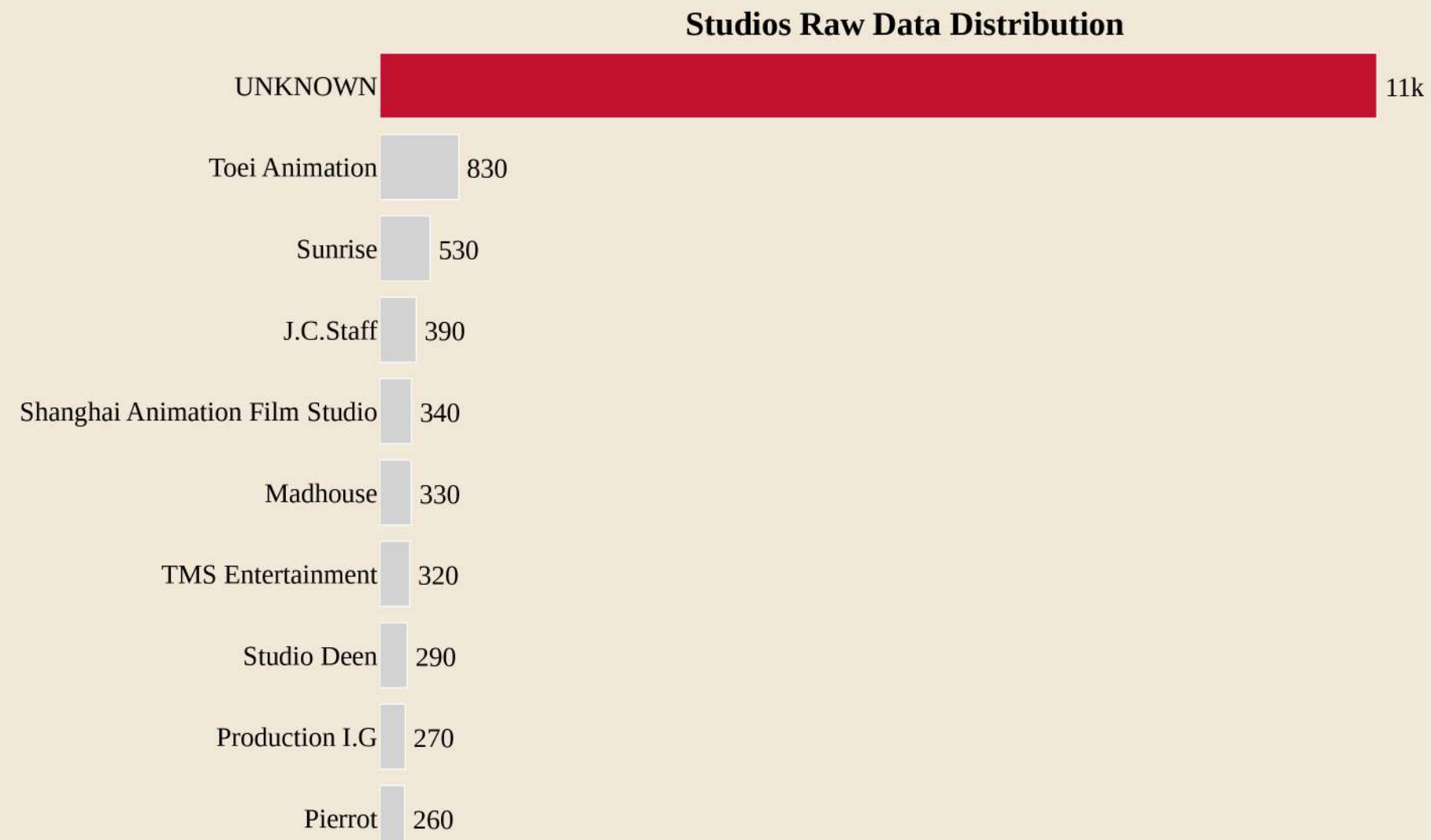


- Same problems with GENRES
- **NHK #1 Producers (820)**
- All producers are **correctly attributed** to the number of invested projects

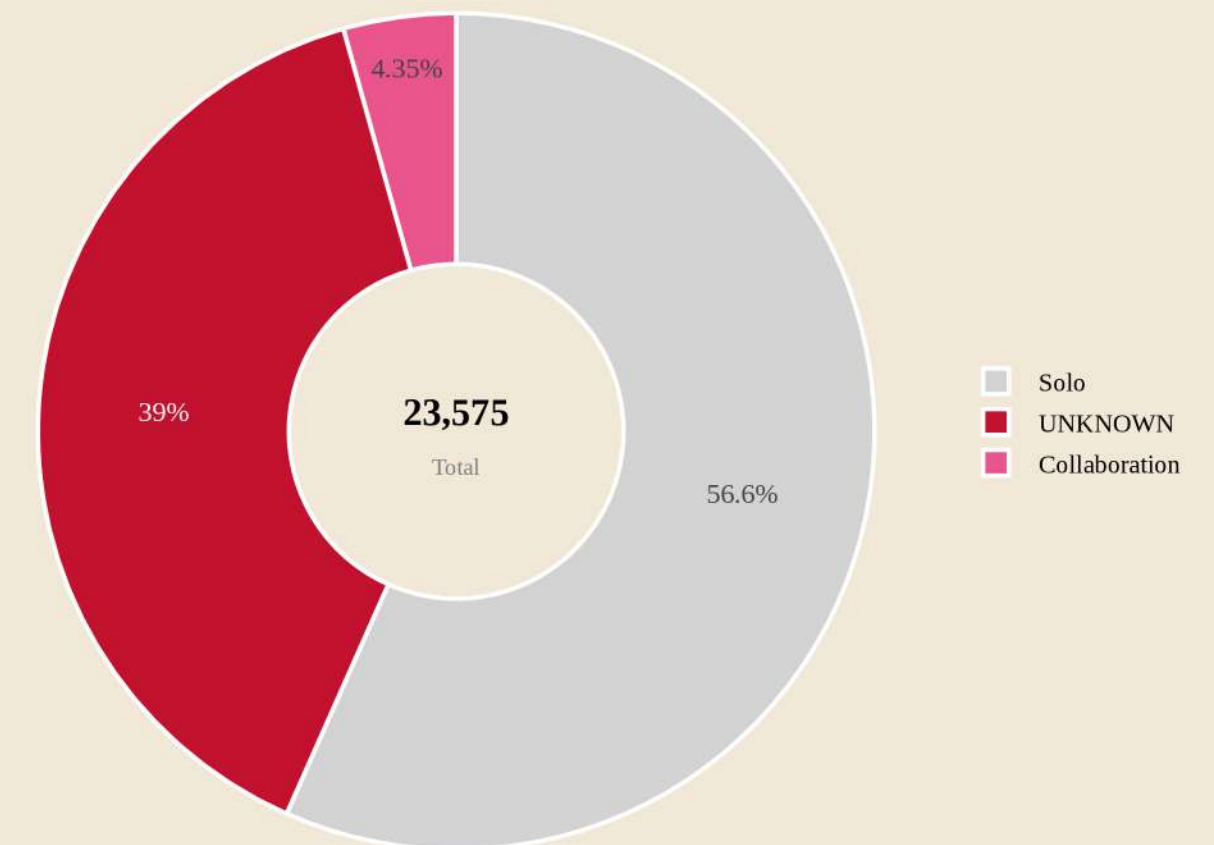
3.3.STUDIOS-RAW DATA CONTAINS UNKNOWN STUDIO

Parsing Studio Data Reveals True Studios Market Leaders

(Top 10 Most Active Studios)



Raw Data: UNKNOWN Dominance and Aggregation In Studio's Proportion



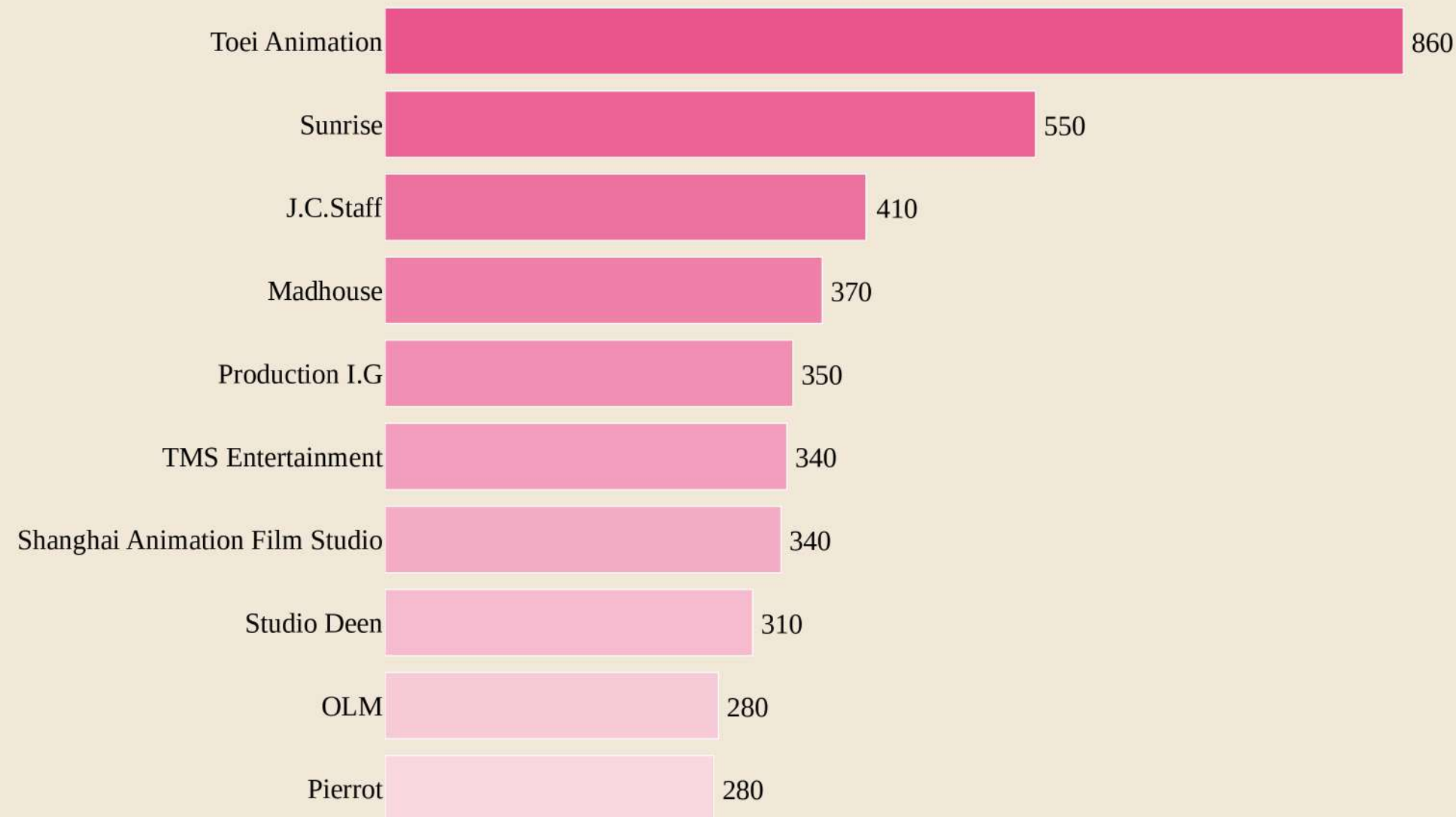
- **UNKNOWN and Solo studios** account for a large portion
- Appearance of **Collaborations**
- **Same problems with GENRES**

3.3.STUDIOS - REVEAL TOP STUDIO IN THE MARKET

Parsing Studio Data Reveals True Studios Market Leaders

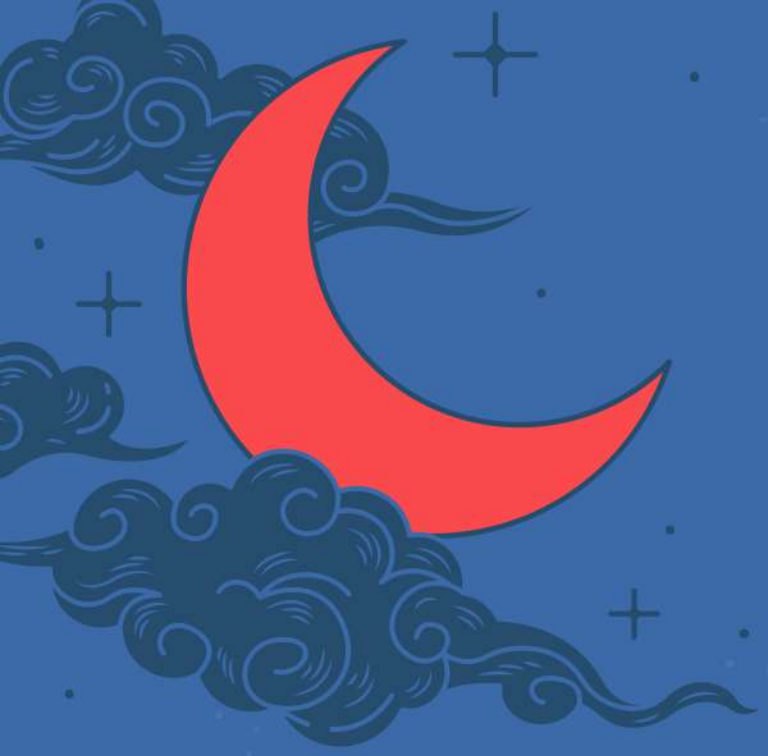
(Top 10 Most Active Studios)

Studios Cleaned Data Distribution



True insight

- Toei Animation #1 Studios (860)
- All studios are **correctly attributed** to the number of invested projects



3.4. BUSINESS INSIGHT

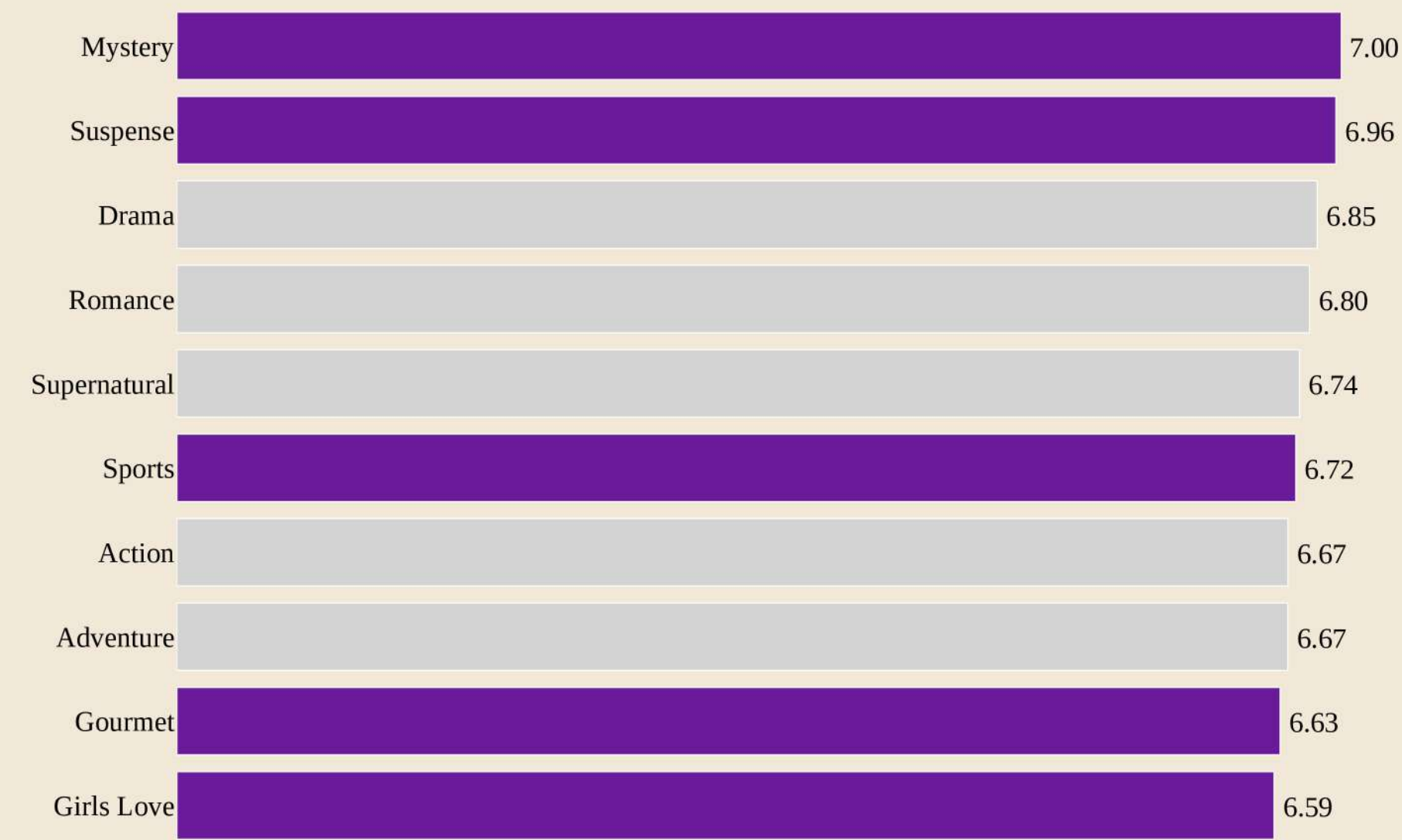
3.4.1. DATA-DRIVEN GENRE INTELLIGENCE

What are the emerging genres with high assessment?

Insight

Niche Genres Dominate Audience Satisfaction Rankings

(Purple: Niche Genres | Grey: Mainstream Genres found in Top 10 Most Aired)



1. **Disaggregation Value:** Multi-label binarization allowed us to isolate specific genre impacts.

→ The data proves that **niche genres** like **Mystery** drive **higher average scores** than **mass-market** genres like **Action**.

2. **Model Signal:** This distinction allows the predictive model to reward 'Quality Niches' rather than just 'Popular Categories'.

3.4.1. GENRES VS SCORE ANALYSIS

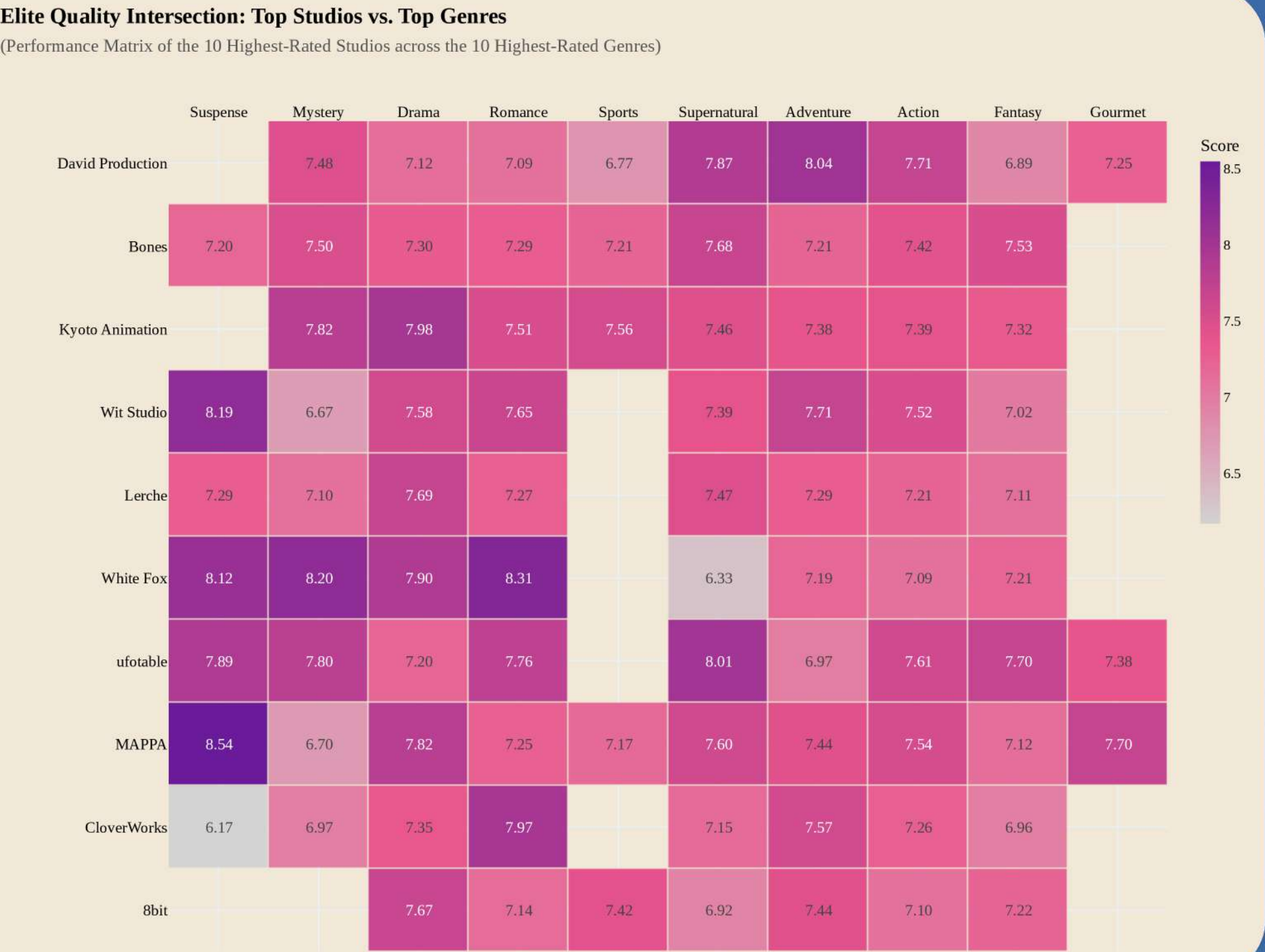
Business Takeaways

- **Max Score Strategy: Prioritize Mystery or Suspense** to chase the highest possible rating.
- **Safe Bet Strategy:** Invest in **Drama or Romance** for a balance of broad appeal and quality.
- **Hybrid Strategy: Create "Super-Niche" titles.** Mix deep Mystery elements with Mainstream themes to get the best of both worlds.



3.4.2. ADVANCED ANALYTICS: STUDIO-GENRE MATRIX

Which Genres do top Studios consistently excel in?



1. Interaction Discovery: This matrix validates the '**Specialist**' hypothesis. Cleaned data confirms that **top studios (e.g., MAPPA, Kyoto Animation) do not perform equally everywhere; they dominate specific niches.**

2. Strategic Value: We transformed raw text into a strategic lookup table, proving that **success depends on Studio-Genre Fit,** not just Studio reputation.

3.4.2. STUDIOS VS GENRES

Business Takeaways

1. Strategy 1: Target the "Triple-A Gold."

- **Focus on the top 3 proven combinations for maximum scores:**
 - MAPPA + Suspense (8.54)
 - White Fox + Romance (8.31)
 - White Fox + Mystery (8.20)

2. Strategy 2: The "Smart Crossover" (Niche Expansion)

- **Leverage core technical skills to dominate less competitive genres:**
 - **Action** → **Sports**: High-motion animation skills transfer perfectly to Sports. (Wit Studio & Bones)
 - **Drama** → **Gourmet**: Strong attention to detail and emotion fits Gourmet storytelling (KyoAni & WhiteFox)

3. Strategy 3: The "Specialist" Rule

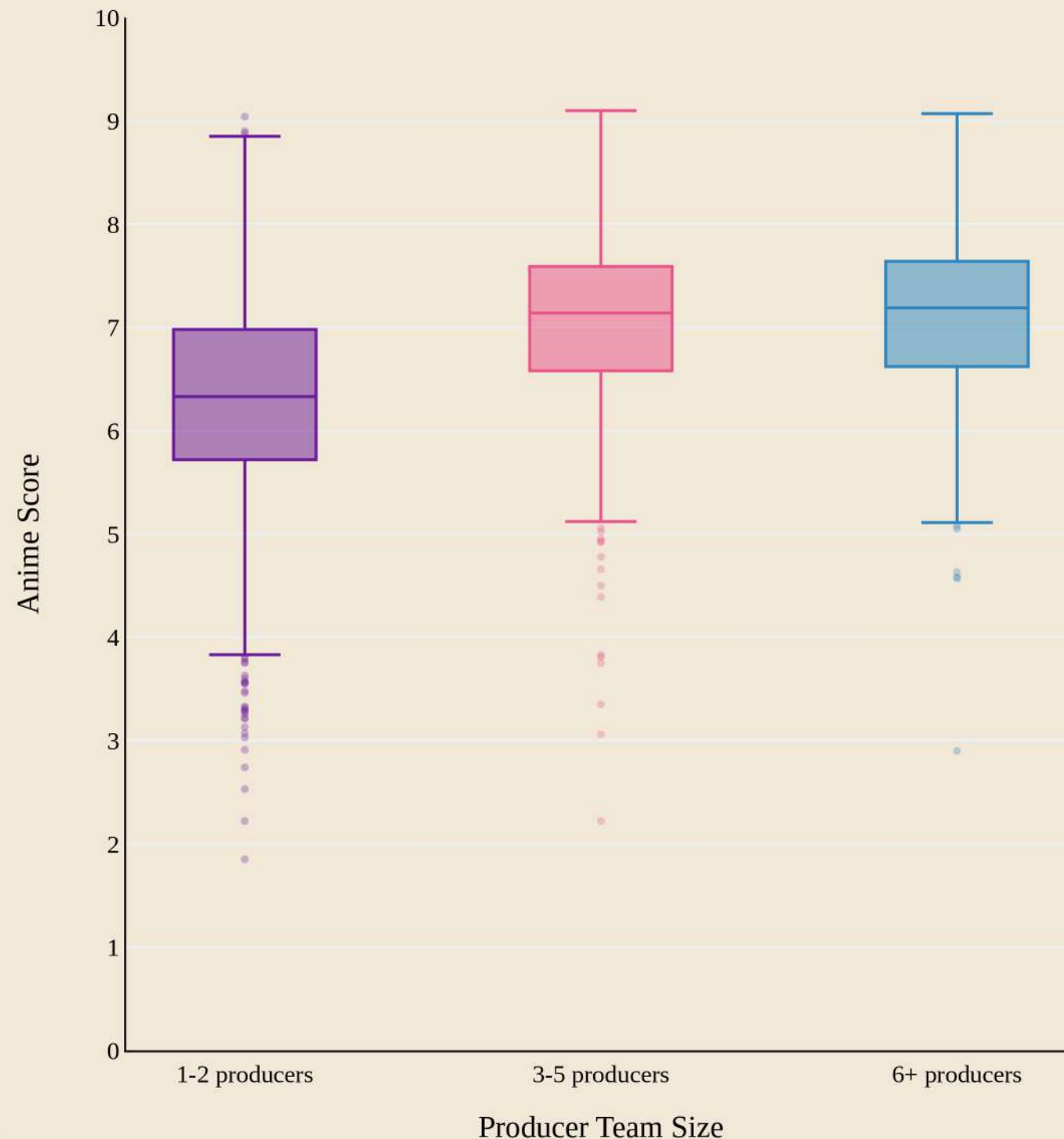
- **To ensure stability and quality, only partner with established leaders:**
 - Adventure: David Production.
 - Supernatural: Ufotable (for high-fidelity spectacle).
 - Drama: Kyoto Animation (The safest bet for consistent quality).

3.4.3. QUANTIFYING THE 'SCALE EFFECT'

Does Committee Size Impact the Score?

Large Collaboration Teams Correlate with Higher Scores

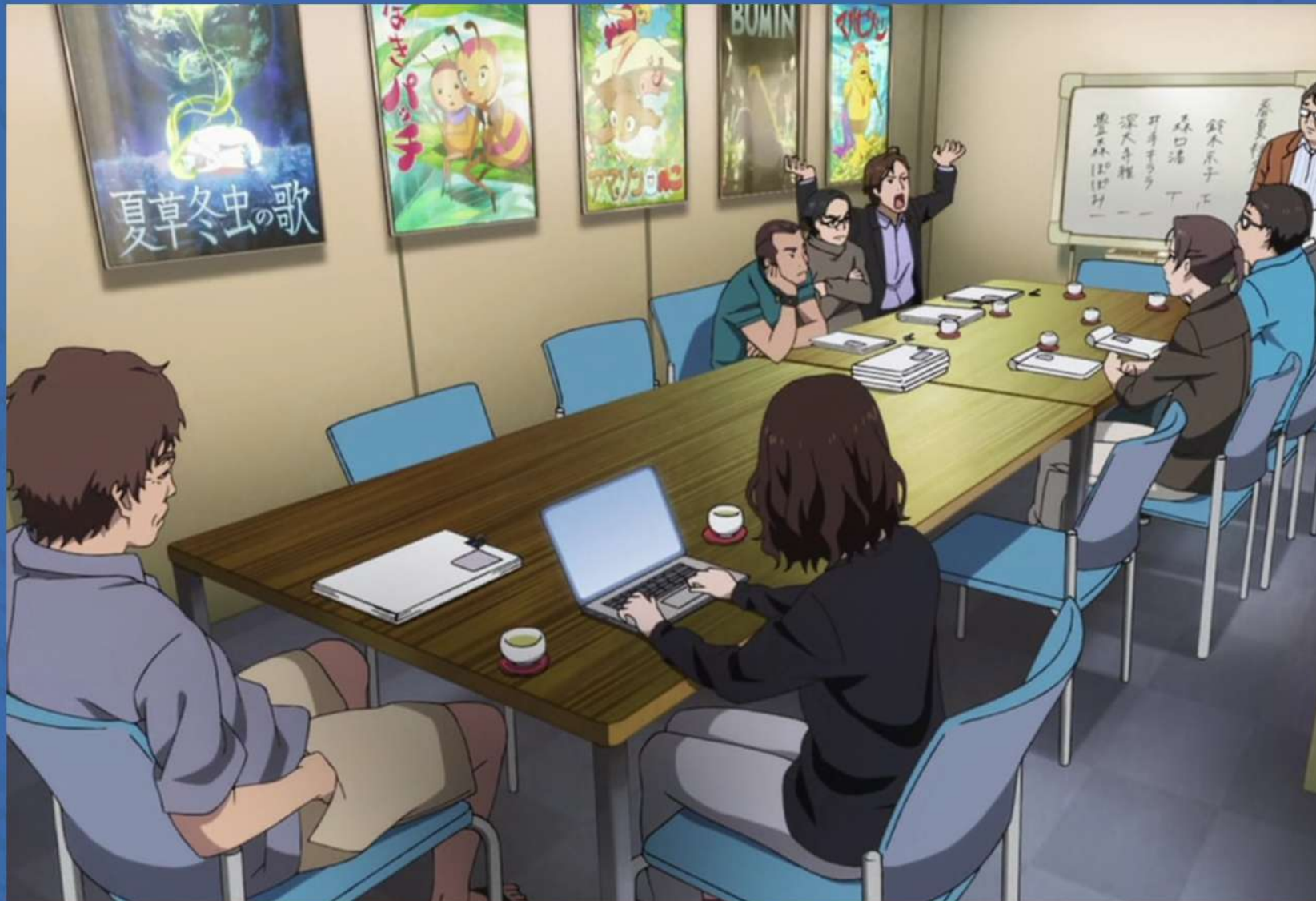
(Avg Score Increases by +0.81 points when comparing 3-5 Producers vs Small Teams)



1. By engineering the 'Committee Size' feature (counting producers), we uncovered a **positive correlation between team size and score.**

2. **Statistical Proof:** The data confirms that **larger committees (3+ partners)** consistently achieve **higher quality floors**, effectively mitigating the risk of failure.

3.4.3. PRODUCERS COMMITTEE



Business Takeaways

- **Optimal Strategy:** Form a Production Committee of **3-5 Partners** as of **diminishing returns**
- **Rationale:** This structure is the "Sweet Spot," **balancing robust financing with management efficiency.**



4. RELEASE STRATEGY

1

Aired Date

2

Episodes

3

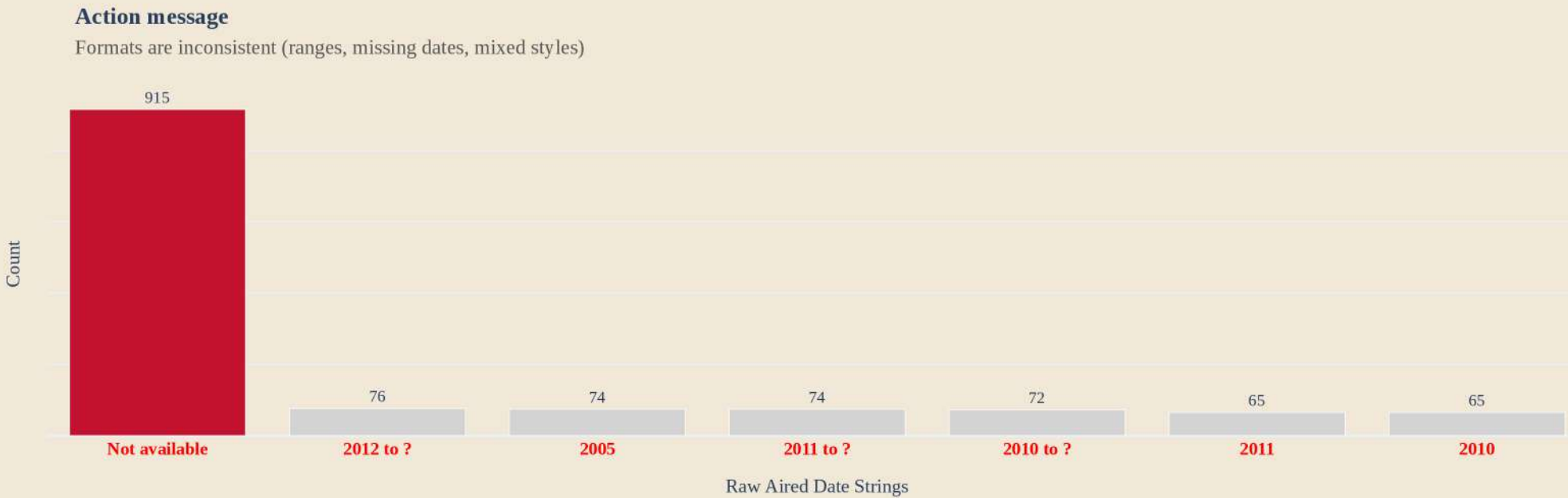
Duration

4.1. FROM UNSTRUCTURED TEXT TO SEASONAL PATTERN

True insight

Raw dates were **messy strings**. Regex parsing allowed us to **extract 'Month'**, revealing **critical seasonal peaks** (Apr, Jul, Oct) that were previously hidden

Raw data



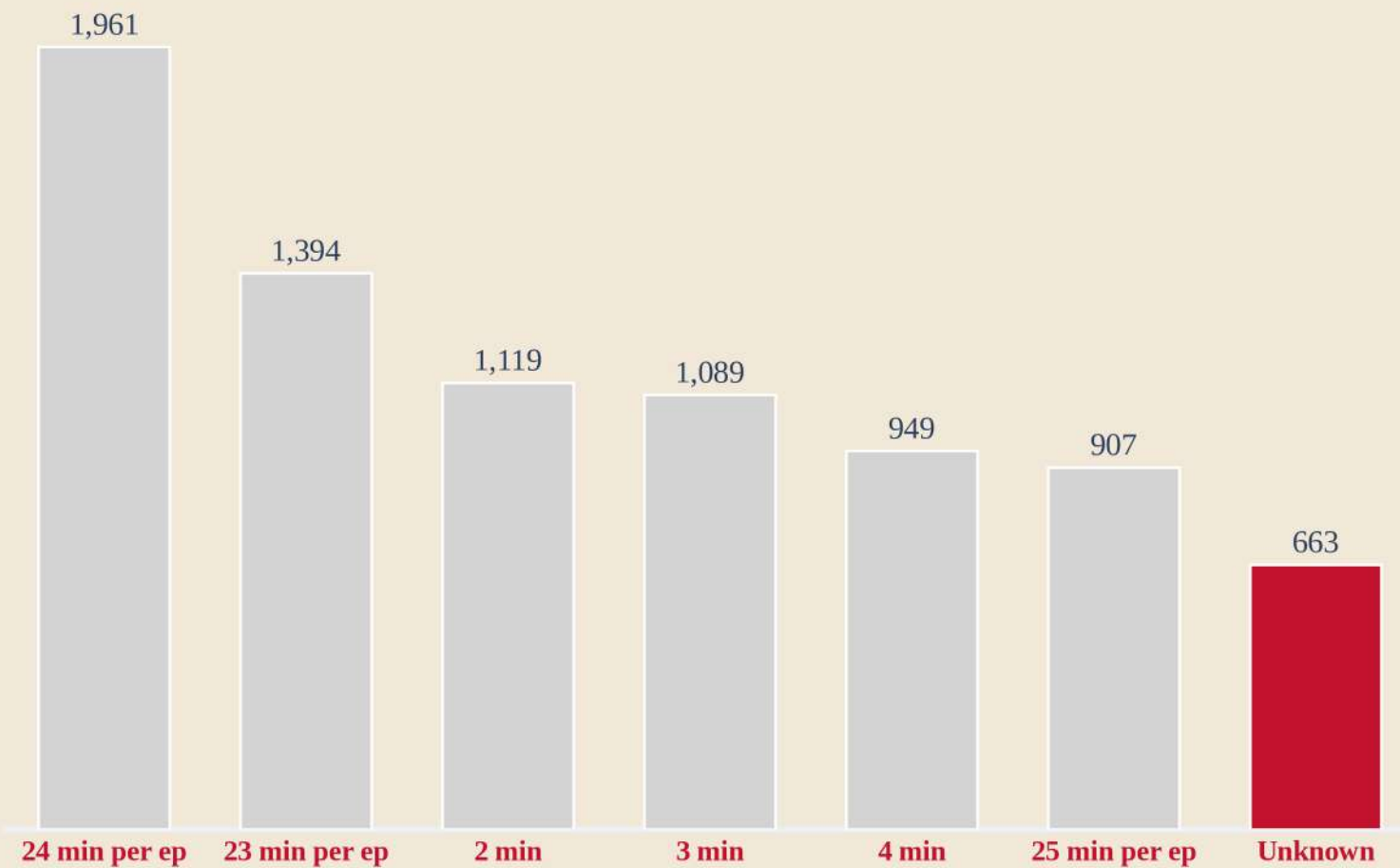
Cleaned data



4.2. UNMASKING THE REAL DURATION DISTRIBUTION

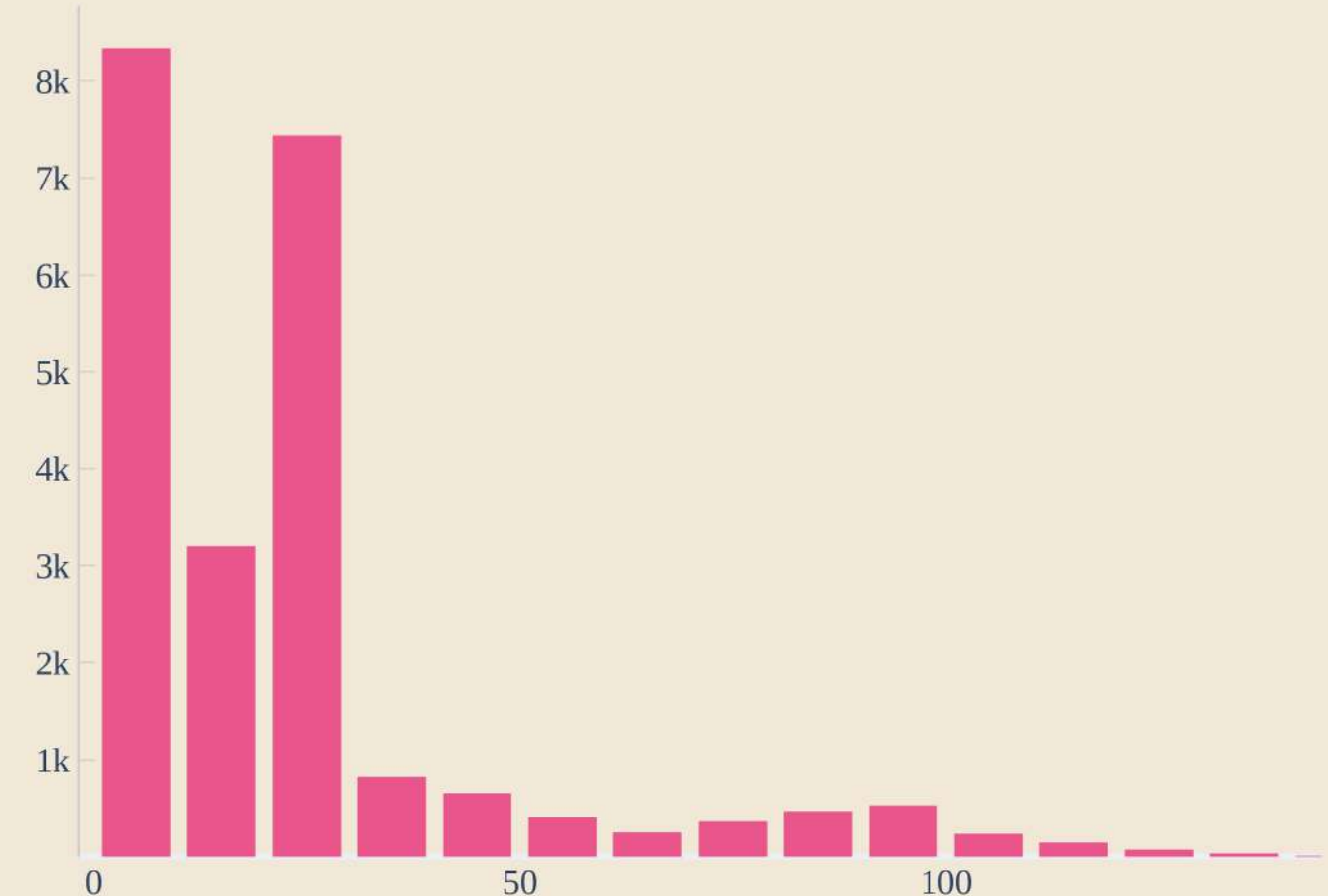
Duration Distribution is Distorted (Raw Data)

(Top 7 Duration Label Frequencies)



True Duration Distribution is Right Skew

(Cleaned Numeric Duration Distribution)



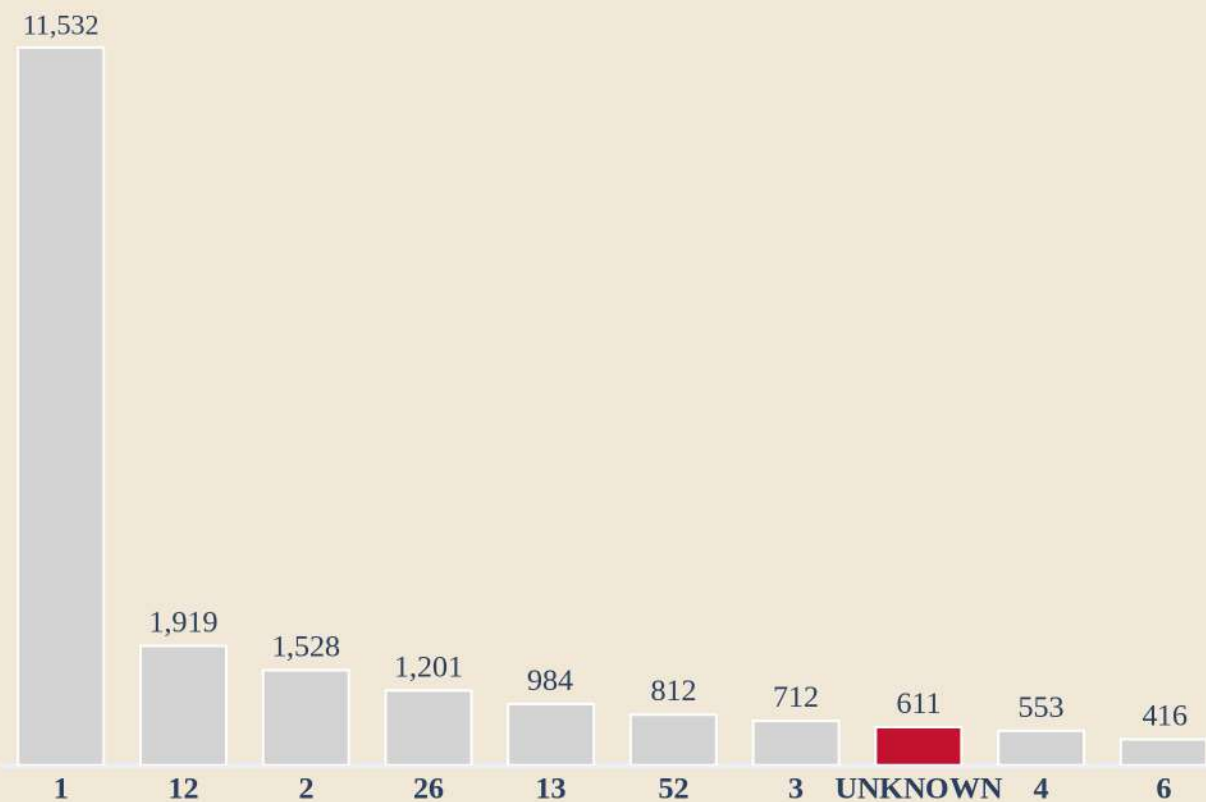
True insight

Raw duration labels were **fragmented and inconsistent**. After cleaning, the data shows a clear **right-skew** with most episodes being **short form**.

4.3. REVEALING THE TRUE EPISODE LENGTH PATTERN

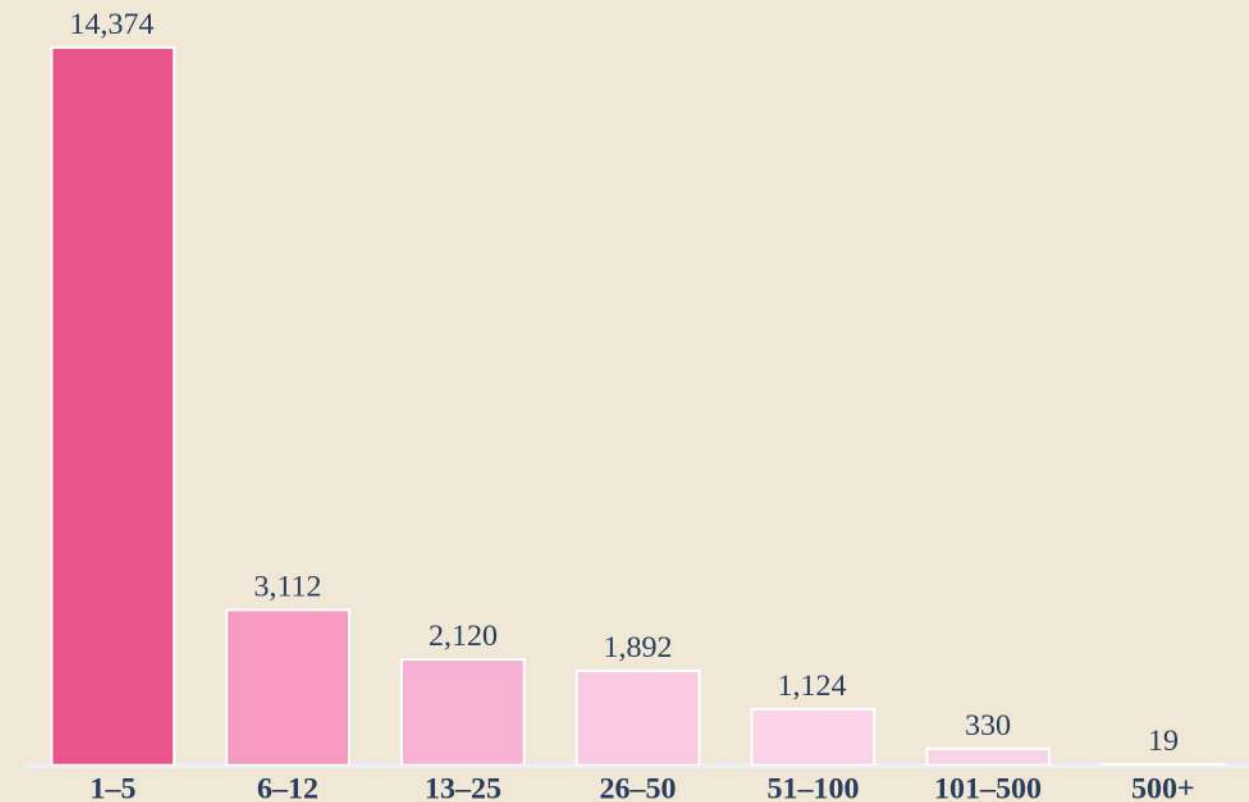
Episode Distribution is Distorted (Raw Data)

(Top 10 Episode Label Frequencies)



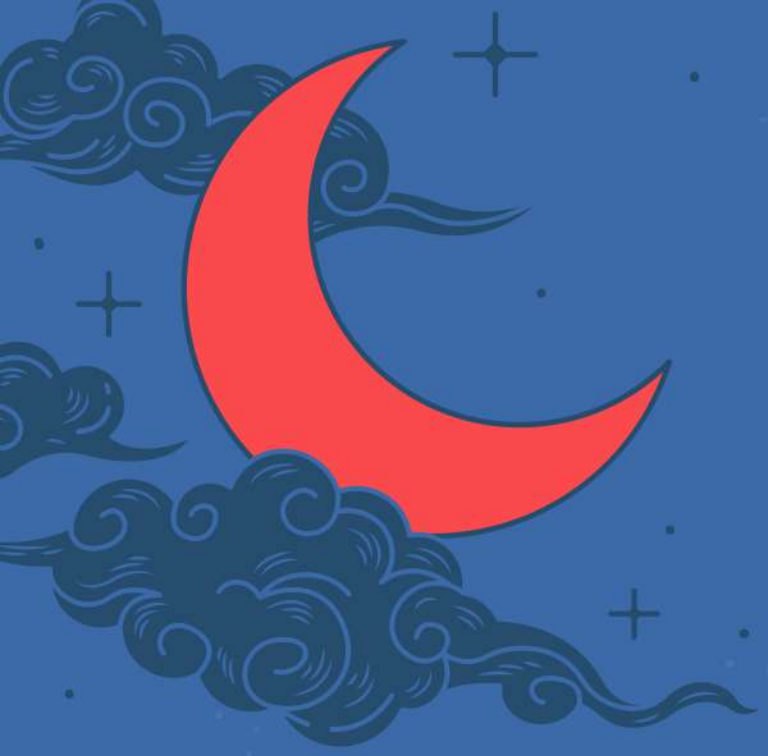
Mini-Series Dominate Anime (Cleaned & Binned)

(Episode Frequencies Across Groups)



True insight

Raw episode were **fragmented** and dominated by “**UNKNOWN**”
After cleaning and binning: most titles are **short mini-series**, with counts dropping sharply as episode length increases.

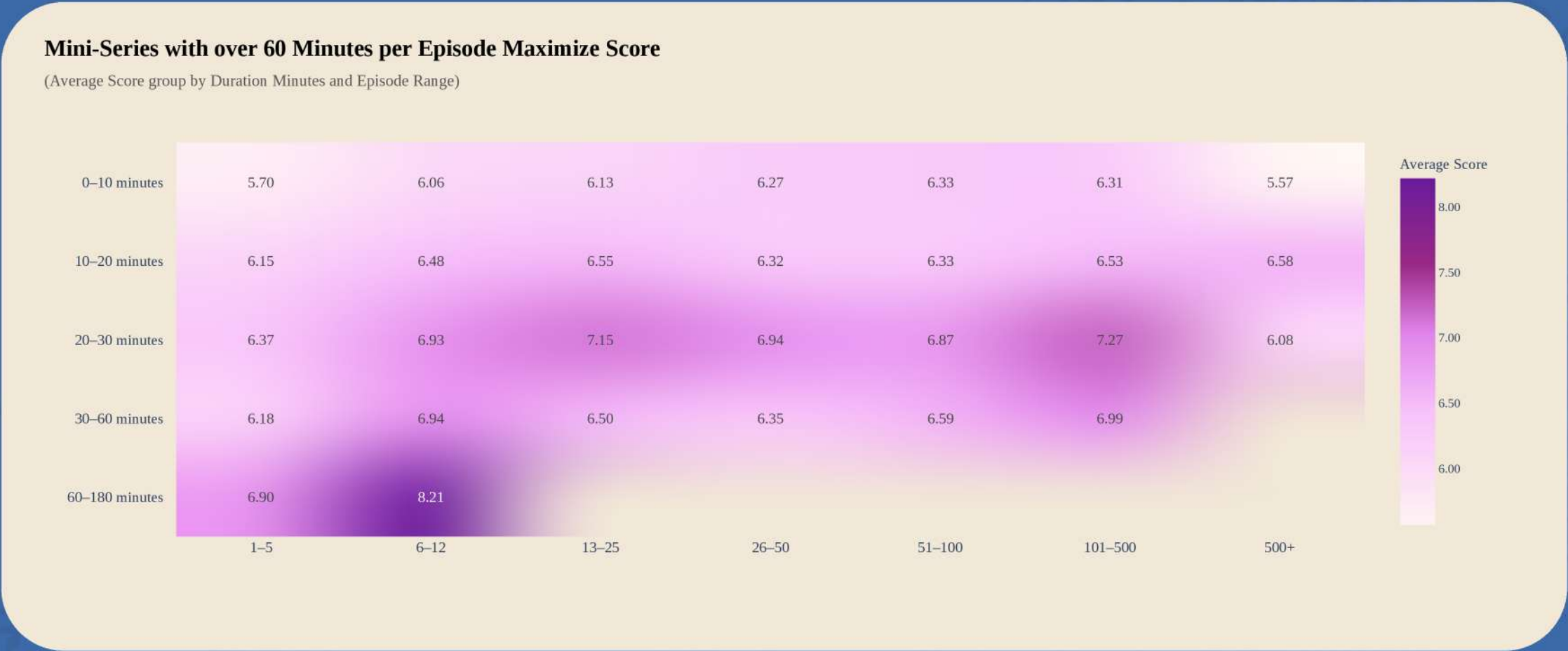


4.4. BUSINESS INSIGHT

4.4.1. FEATURE ENGINEERING REWARD: FINDING THE 'SWEET SPOT'

Insight

- The prepared data confirms that episode count alone doesn't drive quality.
- The cleaned dataset reveals that Mini-series (6–12 eps) achieve the highest scores.
- Feature engineering uncovered that long runtimes matched with few episodes form the true sweet spot.
- Statistical proof shows ultra-short episodes score the lowest.



Business Takeaways

- Prioritize premium limited series with longer runtimes.
- Avoid ultra-short formats as core productions.
- For standard-length episodes, keep series within 12–25 episodes for optimal reception.
- Invest in quality over quantity; pacing and duration drive audience satisfaction.

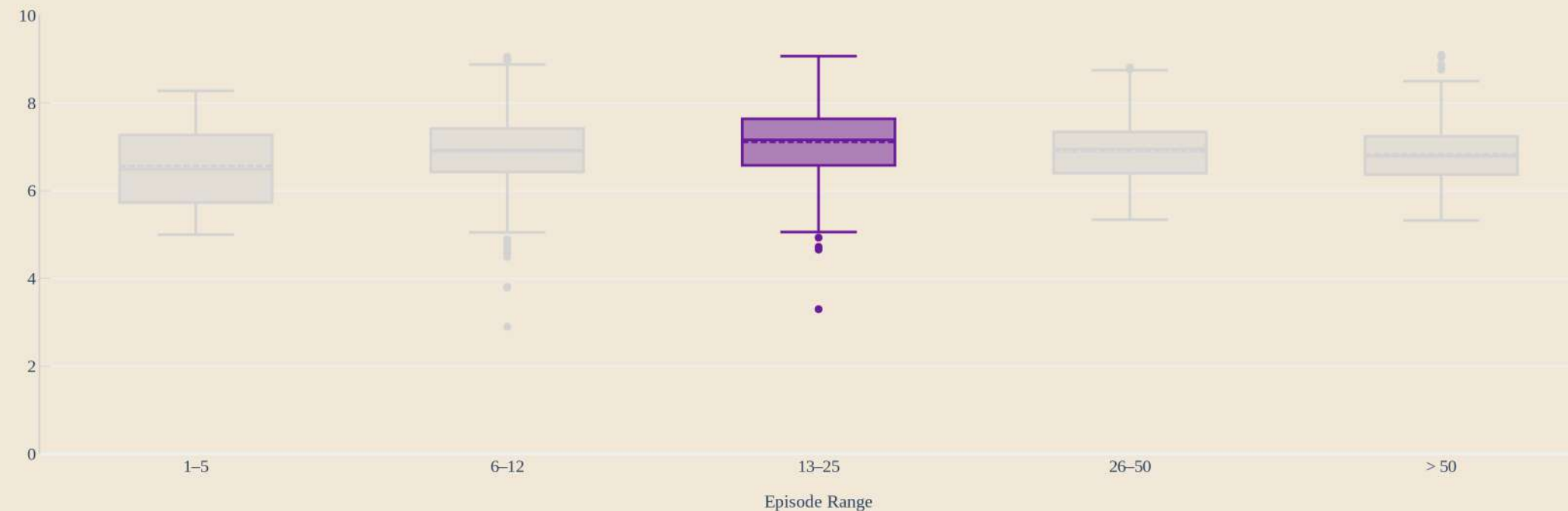
4.4.2. TV ANIME EPISODE STRATEGY

Insight

- Clean data reveals that 13–25 episodes score the highest.
- Duration analysis unlocks how 6–12 episodes fluctuate sharply.
- Patterns validate the stable mid-level of 26–50 episodes.
- Data preparation uncovers how 50+ episodes mask low outliers but yield rare hits.

13–25 Episode TV Series Deliver the Strongest Overall Performance

(Boxplot comparison of Anime TV series score across different episode ranges)



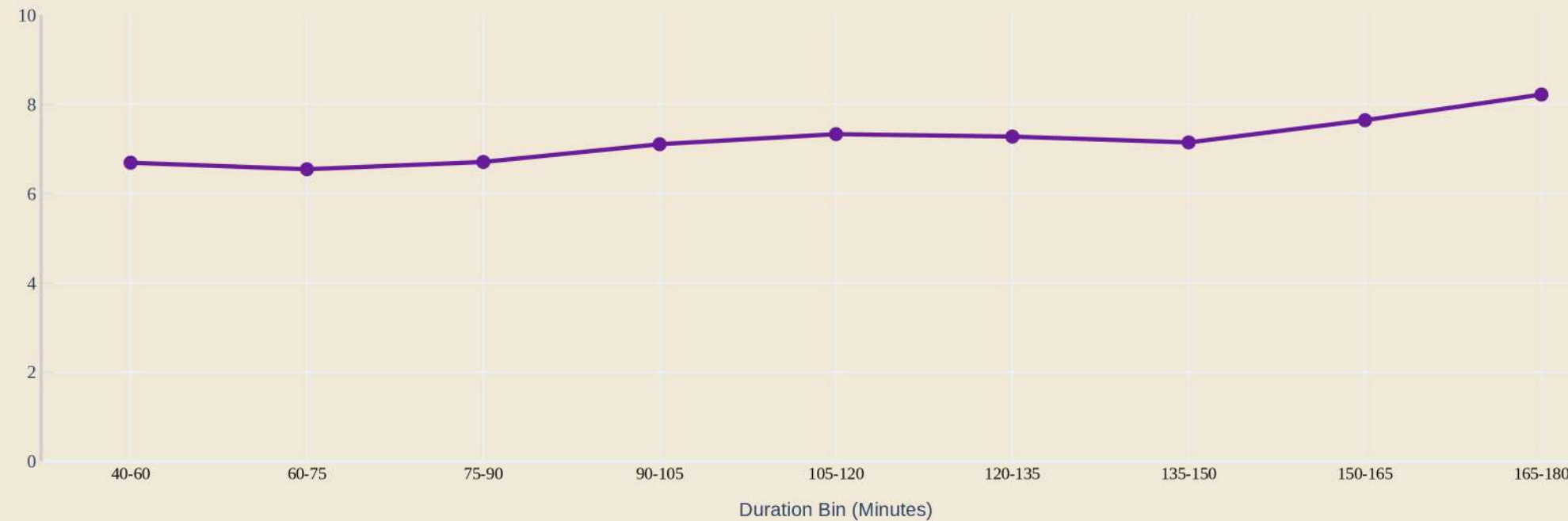
Business Takeaways

- Prioritize 13–25 episode productions for efficient high performance.
- Use 6–12 episode series selectively due to volatility
- Deploy 26–50 episodes for predictable, steady output.
- Treat 50+ episodes as long-term franchise investments with low downside.

4.4.3. VALIDATING UNIT CONSISTENCY

Longer Anime Movies Deliver Higher Average Scores

(Average Score by Duration Bin shows overall upward trend from short to long movies)



Insight

- Normalizing units corrected the distorted distribution, enabling true length–score analysis.
- The cleaned dataset reveals that long films (150–180 min) score the highest.
- Statistical proof shows short films (<90 min) perform the worst.
- The data confirms mid-length films offer stable but not top-tier results.

Business Takeaways

Favor longer movies (150–180 min) for maximum rating potential.

Approach short movies carefully; they generally underperform and require stronger execution.

Mid-length films (90–120 min) offer the safest and most predictable quality.

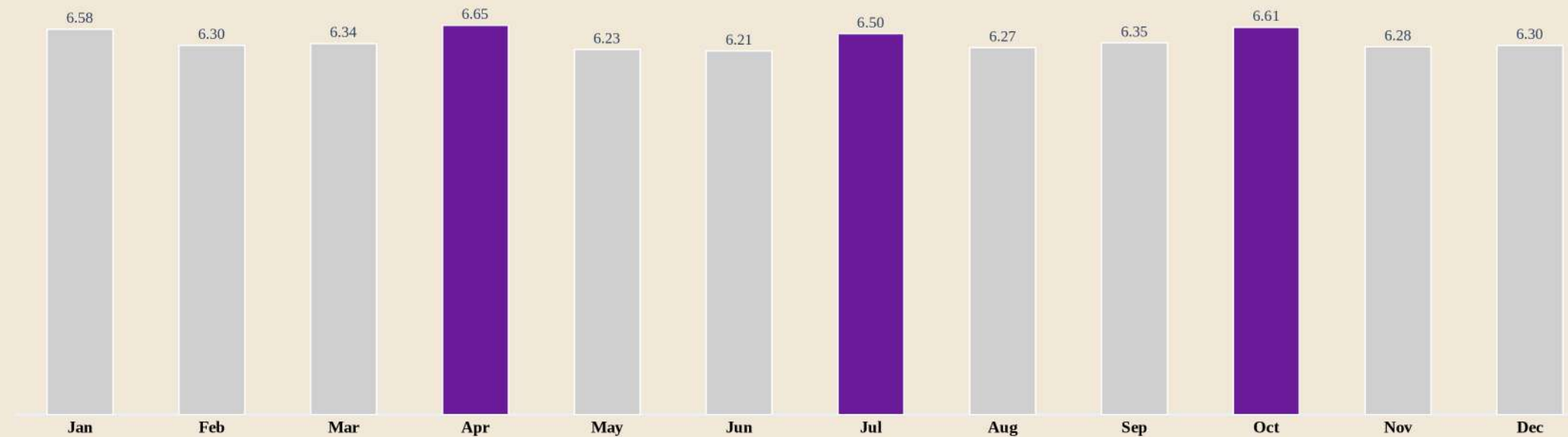
4.4.4. CONFIRMING SEASONAL CYCLES

Insight

- Pattern Recognition: Extracting “Month” from unstructured dates revealed distinct seasonal cycles — with April and October validated as the peak months for high-scoring releases.
- Data Utility: This seasonal signal elevates “Aired Date” from metadata to a predictive timing feature.

Average Anime Score by Month (Clean Data)

Seasonal cycles may influence anime quality



Business Takeaways

Prioritize releasing key titles in **April, July, and October** for maximum impact.

Align marketing pushes with **seasonal peaks** to boost engagement.

Use seasonal patterns to allocate production resources more effectively.

4.4.5. TRACKING HISTORICAL EVOLUTION

Insight

- Long-term Signal: Recovering missing years allowed us to plot the industry's full trajectory. The trend line confirms a steady improvement in production quality from 2000 to the present.
- Baseline Adjustment: This insight helps the model adjust expectations based on the era of release.

Complete Timeline Shows Steady Average Score Improvement

(Average Score by Release Year — Cleaned Data)



Business Takeaways

- Aim for 6.5+ quality to compete in the modern market.
- Prioritize strong storytelling and premium production that aligns with post-2000 audience expectations.
- Reboot older concepts cautiously; legacy ideas require modernization.
- Invest in genres and studios driving the recent score surge (e.g., fantasy, streaming-era hits).

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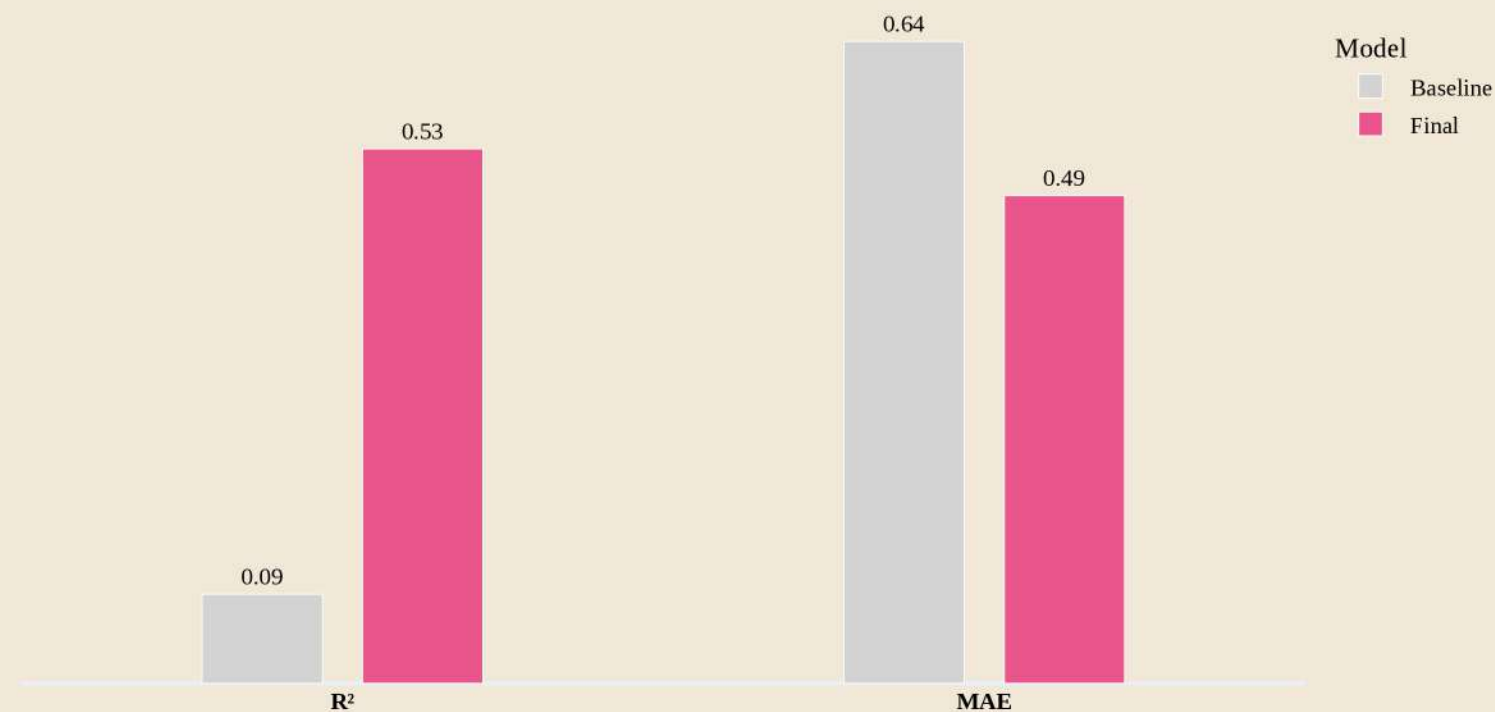
ACT 3

THE RESOLUTION

THE VERDICT: DATA PREPARATION DRIVES PERFORMANCE

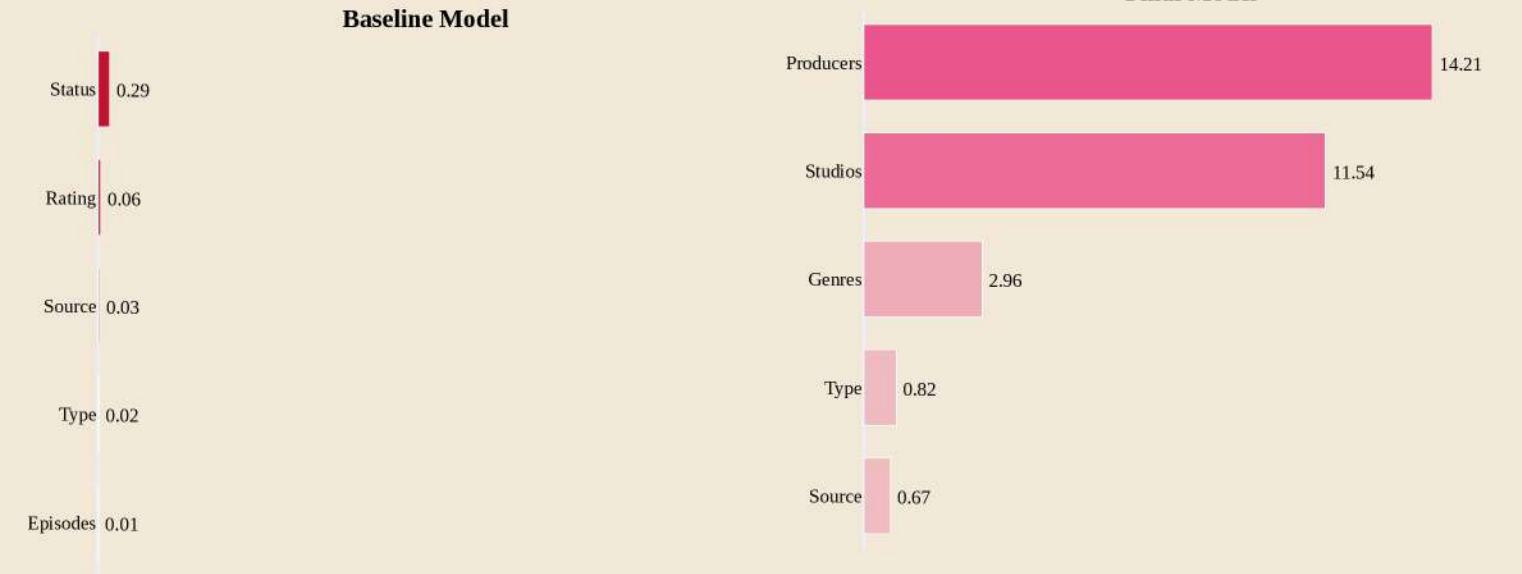
Metrics Before and After Data Preparation

(Metrics Comparison in Baseline and Final Models)



Feature Importance Before and After Data Preparation

(Feature Importance Comparison in Baseline and Final Models)



Metrics

The 6x performance boost is entirely attributable to Data Preparation

The Raw model relied on 'Status' (noise). The Cleaned model correctly identifies 'Producers' & 'Studios' as the true drivers of quality

• DATA-DRIVEN STRATEGIC FRAMEWORK

- **Format Hierarchy:** TV Series and Movies score highest; short-form formats face a natural score ceiling.
- **Source Advantage:** Manga and Novel adaptations outperform Originals and Game-based titles.
- **Studio-Genre Fit:** Top studios excel in specific niches, not across all genres.
- **Scale Effect:** Larger production committees and longer runtimes correlate with higher scores.

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DATA-DRIVEN STRATEGIC FRAMEWORK

- **Core Strategy:** Prioritize TV Series or Movie adapted from high-performing Manga/Novels, supported by large production committees.
- **Targeted Greenlighting:** Invest in high-yield genres like Mystery or Suspense and pair them with specialized studios (MAPPA vs Whitefox)
- **Premium Format Play:** Develop 6–12 episode mini-series and cinematic durations (60–180 min) for the highest score potential.
- **Optimal Timing:** Schedule major releases for Fall or Winter; avoid Summer due to historically lower performance.

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“Raw data deceives. Take action — prepare it and reveal the truth.”

書き初め

A single message that guides the entire story.



NATIONAL ECONOMICS UNIVERSITY
FACULTY OF ECONOMICAL MATHEMATICS



GROUP ASSIGNMENT
The Power of Data Preparation on Anime Dataset
2023 – Technical Report

Group:	01
Members:	Bùi Châu Anh Phạm Vân Thư Thành Uyển Dung Đặng Nhật Huy Chu Bích Phương Vũ Tuấn Đạt
Instructor:	PhD. Nguyễn Tuấn Long
Class:	DSEB 65A

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I. INTRODUCTION AND NARRATIVE FRAMEWORK

1. The Strategic Context

We established a clear objective to anchor the storytelling workflow:

- **The Ambition:** To construct a robust Machine Learning model capable of predicting Anime Scores and identifying the key drivers of audience satisfaction.
- **The Conflict (The Problem):** The raw dataset is not merely "messy"; it suffers from **Structural Entropy** (inconsistent formats), **Critical Blind Spots** (high missingness in key features), and **Signal Distortion** (misleading categorical labels). Using this data "as-is" would lead to "Garbage In, Garbage Out," rendering any predictive model ineffective.
- **The Big Idea:** *"Raw data creates illusions. Only prepared data reveals the truth."* This core message serves as the guiding principle for every transformation step in the pipeline.

2. The Three-Act Structure

We structured the analysis as a coherent narrative arc, guiding the audience through a journey from Diagnostic Confusion to Strategic Clarity. This structure moves beyond a linear series of charts to tell the story of how data quality directly dictates business intelligence.

Act 1: The Context – The "Illusion" of Raw Data

(Corresponding to: Diagnostic Analysis)

- **Narrative Focus:** We begin by establishing the antagonist of our story: the Raw Data itself. Rather than presenting the data as a neutral asset, we frame it as a chaotic force creating "Illusions."
- **The Conflict:** We expose the Strategic Gap defined in our framework. The raw data appears usable on the surface but suffers from Critical Blind Spots (missing information), Structural Entropy (inconsistent formats), and Signal Distortion (misleading categories).
- **Narrative Goal:** To create a sense of urgency. We demonstrate that any decision made (or model built) on this "foggy" foundation would be fundamentally flawed, establishing the necessity for the transformation that follows.

Act 2: The Revelation – Uncovering the Truth

(Corresponding to: Comparative Analysis)

- **Narrative Focus:** The Discovery. This is the core of our story, where we peel back the layers of noise to reveal the true market dynamics. We organized this act into four thematic pillars to maintain a logical flow:
 - **The Foundation:** Moving from unstructured text to a measurable target variable.
 - **Market Factors:** Revealing the true market leaders by resolving "Unknown" ambiguities.

- **Creative Factors:** Determining the actual impact of Studios and Genres by deconstructing fragmented lists.
- **Release Strategy:** Unlocking the hidden relationship between time, duration, and audience reception.
- **Narrative Goal:** To execute a "Compare and Contrast" strategy. By contrasting the "Raw Illusion" against the "Cleaned Truth," we visually prove that the insights are not random—they are the direct result of rigorous preparation.

Act 3: The Resolution – The Verdict & Call to Action

(Corresponding to: Model Validation & Conclusion)

- **Narrative Focus:** The Proof. We move from visual intuition to mathematical certainty. This act validates that our "Cleaned Truth" is not just aesthetically pleasing but statistically robust.
- **The Resolution:** We present the dramatic improvement in model performance as the definitive "Verdict," confirming that the noise has been successfully converted into a predictive signal.
- **Call to Action:** The story concludes by synthesizing the uncovered insights into a **Data-Driven Strategic Framework**. We end with a definitive directive: Stakeholders must shift from intuition-based guesses to evidence-based precision enabled by prepared data.

3. Applied Storytelling Techniques

To ensure the report is persuasive and easy to follow, we applied specific principles from *Storytelling with Data*:

- **Cognitive Load Reduction:** We deliberately separated the heavy engineering code (Notebook 02) from the visual storytelling (Notebook 03) to keep the audience focused on insights, not syntax.
- **Bing, Bang, Bongo:**
 - Bing (Introduction): We explicitly defined the "Strategic Gap" and the three main data issues (Blind Spots, Entropy, Distortion) upfront.
 - Bang (Analysis): We demonstrated the resolution of these exact issues through detailed "Before vs. After" deep dives in the Comparative Analysis section.
 - Bongo (Conclusion): We synthesized the findings into a Strategic Framework and validated the initial claim with the final model performance metrics, closing the loop with a definitive Call to Action.
- **Vertical Logic:** Each section in Act 2 follows a consistent structure: *Identify Issue -> Technical Solution -> Visual Evidence -> Business Insight (Reward)*. This repetition helps the reader follow the logic effortlessly.
- **The "Compare and Contrast" Tactic:** By placing "Raw Data" and "Clean Data" charts side-by-side (using Subplots), we force the audience to see the immediate value of the data preparation process.

II. THE ENGINEERING WORKFLOW

To bridge the gap between raw data and actionable insights, we executed a three-stage engineering workflow. This process transformed chaotic information into a structured asset for both visualization and modeling.

1. Stage 1: Diagnostic Assessment (The Health Check)

Reference: Notebook 01

We started by auditing the raw dataset to identify why it could not be used immediately. The diagnosis revealed that the data was structurally broken: key metrics like Score and Duration were trapped in inconsistent text formats, and categorical fields contained too much noise to be analyzed effectively.

2. Stage 2: Structural Cleaning (Preparing for Discovery)

Reference: Notebook 02 (Part 1)

Before visualizing any trends, we had to standardize the data format.

- **The Action:** We implemented a comprehensive cleaning strategy utilizing Regular Expressions (Regex) to parse complex text strings into usable numbers (e.g., extracting minutes from *Duration*, parsing years from *Aired* dates). Furthermore, we applied Business Logic Validation to correct semantic errors—such as fixing *Episode* counts for Movies, resolving *Status* conflicts—and performed Macro-Categorization to consolidate fragmented *Sources*, among other structural rectifications.
- **The Goal:** To create a "clean" dataset (`prepared_data.csv`) that allows for accurate **Comparative Analysis** in Act 2.

3. Stage 3: Automated Transformation (Preparing for Modeling)

Reference: Notebook 02 (Part 2)

To enable Machine Learning, we built an automated processing pipeline.

- **The Action:** We engineered Custom Transformers to specifically handle complex multi-label list data (like *Genres* and *Studios*) that standard tools cannot process. The pipeline also includes Dimensionality Reduction to filter out noise from rare categories, Feature Engineering (such as binning continuous variables, creating interaction terms), and cyclical encoding for temporal features.
- **The Goal:** To produce a refined feature set that maximizes the predictive power of our model in Act 3.

III. VISUALIZATION DESIGN SYSTEM

To ensure the technical analysis is communicative rather than just descriptive, we established a strict design system based on the principles of Data Visualization. This system governs every chart in the notebook, ensuring consistency and reducing cognitive load.

1. Tooling & Technical Environment

We utilized **Plotly (Python)** as our primary visualization engine.

- **Rationale:** Unlike static libraries (Matplotlib), Plotly allows for granular control over every chart element via `graph_objects` and the interactive layout. This was essential for implementing custom layouts (Subplots), removing default clutter, and exporting

high-resolution vector graphics (SVG/PNG via Kaleido) suitable for professional reporting.

2. Decluttering Strategy (Gestalt Principles)

We maximized the **Data-to-Ink Ratio** by systematically removing non-essential elements, leveraging Gestalt principles to guide the viewer's cognitive processing:

- **Law of Proximity:** In comparative analysis, we placed "Raw" and "Clean" charts within the same subplot coordinates. This physical closeness forces the brain to perceive them as a single comparative unit rather than two separate entities.
- **Strategic Gridlines:** We removed distracting gridlines in simple categorical comparisons (like Bar Charts) where the shape of data matters more than the exact number. However, we retained specific reference lines (e.g., Y-axis grids in Box Plots) where precise value comparison was critical.
- **Direct Labeling:** We replaced peripheral legends with direct text labels on bars and lines wherever possible. This prevents the "eye-scanning fatigue" caused by looking back and forth between the data and a legend box.

3. Preattentive Attributes (Semantic Color Logic)

Color was not used for decoration but as a **functional code**. We defined a semantic palette to direct attention to the narrative conflict and resolution:

-  **Alert Red (#C3122F) – The "Antagonist":**
 - *Usage:* Assigned exclusively to Missing Values, "Unknown" labels, and Outliers.
 - *Rationale:* Red biologically triggers immediate attention and is culturally associated with "Stop" or "Warning." This highlights the *Data Chaos* (Act 1).
-  **Neutral Grey (#D3D3D3) – The "Context":**
 - *Usage:* Used for background categories, non-top-10 items, or raw data comparison baselines.
 - *Rationale:* Low saturation pushes these elements to the visual background, ensuring they provide context without competing for attention.
-  **Vibrant Pink (#ea568c) &  Deep Purple (#6A1B9A) – The "Protagonist":**
 - *Usage:* Pink denotes Valid/Cleaned Data (the corrected state). Purple highlights Strategic Opportunities (e.g., High-scoring Niche Genres, Top Studios).
 - *Rationale:* We utilized these hues to establish a visual hierarchy of "Truth." While Red captures attention for *Data Defects* (Noise), Pink and Purple represent the "Recovered Signal"—the actionable insights that were previously hidden. This creates a clear visual contrast: Red is what we must *fix*, while Pink/Purple is what we must *focus on* for decision-making.

4. Standardization & Typography

To maintain a professional academic standard across the report, we enforced rigid formatting rules:

- **Typography:** All text elements utilize the **"Tahoma"** font family. This sans-serif font ensures high readability on digital screens and aligns with modern design standards.
- **Action Titles:** Every chart features a two-tier title structure:
 - *Primary Title (Bold):* States the **Insight** (e.g., *"Mini-Series Maximize Score"*).
 - *Subtitle (Grey, Smaller):* Describes the **Data** (e.g., *"Average Score by Episode Count"*).
- **Number Formatting:** We standardized granularity (e.g., scores to 2 decimal places, counts with 'k' suffixes) to reduce numerical clutter.

IV. DIAGNOSTIC VISUALIZATION ANALYSIS (ACT 1)

To visually justify the need for extensive data preparation, we selected two diagnostic charts that expose the critical flaws within the raw dataset. These visualizations serve as the "Evidence of Conflict" in our narrative arc.

1. Case Study 1: Auditing Data Completeness

Columns with high missing data require cleaning

(Missing data ratio by column)

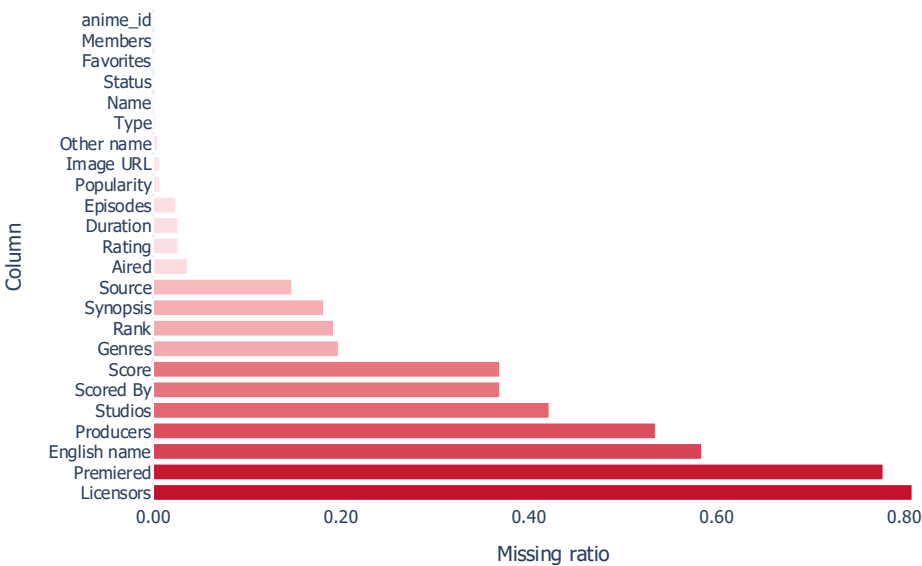


Figure 1: Systemic Data Gaps in Business-Critical Columns.

a. Visual Rationale

We employed a Horizontal Bar Chart to audit the integrity of the dataset columns. A horizontal orientation was chosen over a vertical one to accommodate long feature names (e.g., "English name", "Licensors") without requiring rotation, ensuring immediate readability.

b. Decluttering Strategy

- **Sorted Hierarchy:** The bars are strictly sorted by missingness ratio (Descending). This reduces cognitive load by allowing the viewer to instantly identify the "worst offenders" at the bottom without scanning the entire chart.
- **Direct Comparison:** By normalizing the count to a ratio (0.0 to 1.0), we make the scale intuitive regardless of the total dataset size.

c. Preattentive Attributes

We utilized a Sequential Red Palette as a warning signal:

- **Intensity as Severity:** The color deepens from light pink (low missingness) to dark red (high missingness). This preattentive attribute draws the eye directly to the bottom of the chart—specifically to Licensors (~80% missing) and Premiered (~78% missing).
- **Semantic Color Association:** Red is universally associated with "Stop" or "Error," reinforcing the message that these specific columns are currently unusable for modeling.

d. The Resulting Insight

The visualization proves that Monetization Data (Licensors) and Seasonal Data (Premiered) are fundamentally broken. Any attempt to use these features in the model without aggressive cleaning or imputation would result in failure.

2. Case Study 2: Inspecting Target Variable Health

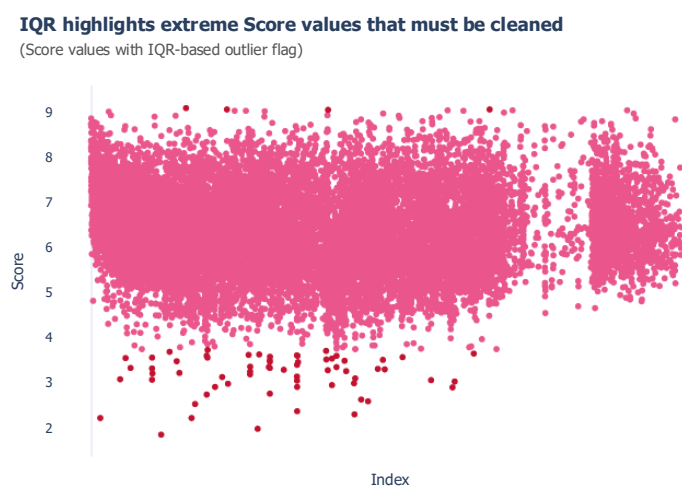


Figure 2: Target Variable (Score) Compromised by Statistical Noise.

a. Visual Rationale

To assess the quality of our target variable (Score), we selected a Scatter Plot over a simple Histogram. While a histogram shows distribution shape, a scatter plot reveals the density and

specific location of anomalies relative to the index, allowing us to see if errors are clustered or random.

b. Decluttering Strategy

- **Minimalist Axes:** We removed heavy gridlines to focus solely on the data points.
- **Binary Classification:** Instead of plotting raw values indiscriminately, we calculated the Interquartile Range (IQR) beforehand and flagged data points as either "Normal" or "Outlier."

c. Preattentive Attributes

- **Strategic Contrast:** We used a complementary color scheme to separate signal from noise.
 - **Pink (Bulk Data):** Represents the valid data points, forming the "body" of the distribution.
 - **Deep Red (Outliers):** Represents extreme values (statistically implausible scores given the context). The distinct contrast separates the "Mainstream" anime from the "Edge cases."

d. The Resulting Insight

The chart reveals a dense "cloud" of valid scores between 6.0 and 8.0, but it is plagued by a significant trail of low-score outliers (red dots below 4.0). These outliers represent "trash" data or statistical anomalies that would skew a Linear Regression model (which is sensitive to means).

V. COMPARATIVE VISUALIZATION ANALYSIS (ACT 2)

To validate the impact of our data preparation pipeline, we analyze seven distinct visualization scenarios. Each case represents a specific class of data problem—from distribution errors to multivariate complexity—demonstrating how our design choices transformed raw noise into strategic intelligence.

1. Case Study 1: Transforming Target Variable Distribution

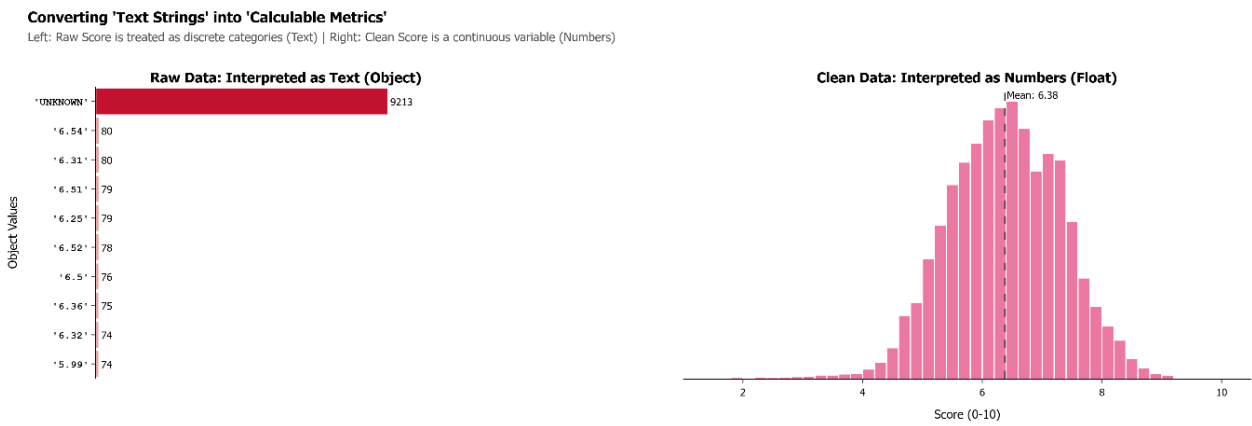


Figure 3: Visualizing the Shift from Categorical Noise to Numerical Distribution

a. Visual Rationale

Instead of using a generic table or summary statistics, we deliberately paired two distinct chart types to tell the "Before & After" story:

- **Left (Raw Data):** We selected a Horizontal Bar Chart because the raw data was categorical (strings). A horizontal layout allows for long labels (like specific text anomalies) to be read naturally without head-tilting, which is superior to a vertical column chart for categorical data.
- **Right (Clean Data):** We switched to a Histogram because the goal changed from "identifying errors" to "analyzing distribution." A histogram is the only valid choice to visualize the shape (Bell curve), spread, and central tendency of continuous numerical data.

b. Decluttering Strategy

Following the "Data-to-Ink Ratio" principle, we stripped away non-essential elements to force the viewer to focus solely on the data comparison:

- **Removed Legends:** Since the title and subtitles clearly define "Raw" vs. "Clean," a separate legend box would be redundant clutter. We removed it to save whitespace.
- **Simplified Axes:** We removed the heavy borders and background shading (using `plotly_white`). On the right chart, we reduced the prominence of X-axis ticks to emphasize the *shape* of the distribution rather than specific granular values.
- **High-Contrast Layout:** By using a transparent background and white space, the colored bars become the only visual weight, eliminating "chart junk."

c. Preattentive Attributes

We used visual cues that the brain processes in milliseconds (before conscious reading):

- **Color as Status:** We avoided generic blue. Red (#C3122F) was assigned to the "UNKNOWN" bar to instantly signal a "Stop/Warning" state. Pink (#ea568c) was used for the clean chart to signal a "Valid/Safe" state.
- **The "Mean" Anchor:** The vertical dashed line on the right chart acts as a visual anchor. It draws the eye immediately to the center (6.38), proving physically that the data now has a calculable center—something impossible in the left chart.

d. The Resulting Insight

This comparison provides immediate visual proof of the engineering success. It demonstrates that the chaotic text data (dominated by 9,213 "Unknown" entries) has been successfully transmuted into a Gaussian-like distribution centered at 6.38. This visually validates that the target variable is now statistically sound and ready for the regression modeling.

2. Case Study 2: Consolidating Market Fragmentation

Consolidating Fragmented Sources Reveals True Market Structure

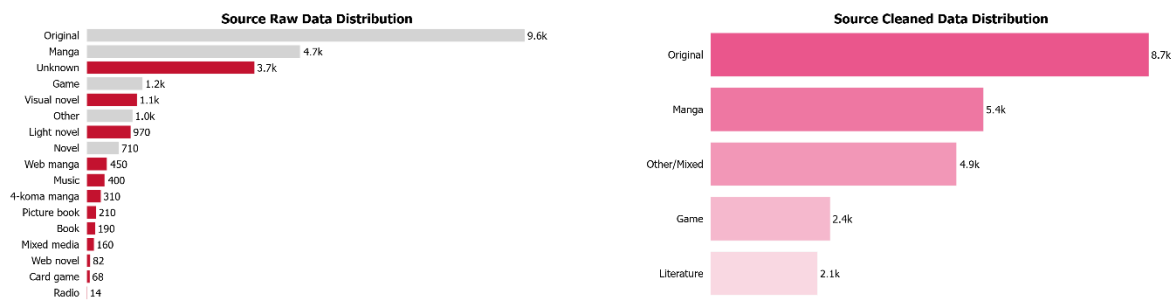


Figure 4: Consolidating Fragmented Sources Reveals True Market Structure

a. Visual Rationale

We faced a problem of fragmentation: the raw data contained redundant micro-categories (e.g., *Web manga*, *4-koma manga*) that diluted the visibility of major market segments.

- **Paired Subplots:** We chose a side-by-side layout (1 row, 2 columns) to force a direct comparison between the "Fragmented" state and the "Consolidated" state.
- **Horizontal Orientation:** Since category names like "*Light novel*" or "*Visual novel*" are long text strings, a vertical bar chart would require rotating labels (violating readability rules). Horizontal bars allow the labels to be read naturally from left to right.

b. Decluttering Strategy

We maximized the "Data-to-Ink Ratio" by removing elements that forced unnecessary eye movement:

- **Direct Labeling:** Instead of forcing the viewer to scan down to an X-axis to guess the value, we placed the exact counts (e.g., "9.6k", "5.4k") directly at the end of each bar. This allowed us to remove the X-axis gridlines entirely, leaving a clean white background.
- **Visual Aggregation:** The cleaning process itself was a form of decluttering. By reducing the number of bars in the right chart (grouping micro-types into Macro-types), we physically lowered the cognitive load required to process the market structure.

c. Preattentive Attributes

We used color strategically to narrate the data quality transformation:

- **Red as Warning:** In the Raw chart (Left), we assigned Red (#C3122F) to the massive "Unknown" bar (3.7k) and fragmented categories. This draws the eye immediately to the "mess" that needs fixing.
- **Gradient as Hierarchy:** In the Clean chart (Right), we moved to a Sequential Pink Gradient. The color intensity scales with the volume (Darker = Higher Count). This naturally guides the viewer's eye to the top contributors (Original and Manga) without needing extra arrows or bold text.

d. Resulting Insight

The visualization confirms that while "Original" remains the top source (~8.7k), the grouping strategy successfully recovered "Manga" as a major pillar (growing from 4.7k to 5.4k) and established "Literature" as a clear segment, proving that the raw data significantly under-represented these key formats.

3. Case Study 3: Ranking with Visual Hierarchy

Niche Genres Dominate Audience Satisfaction Rankings

(Purple: Niche Genres | Grey: Mainstream Genres found in Top 10 Most Aired)

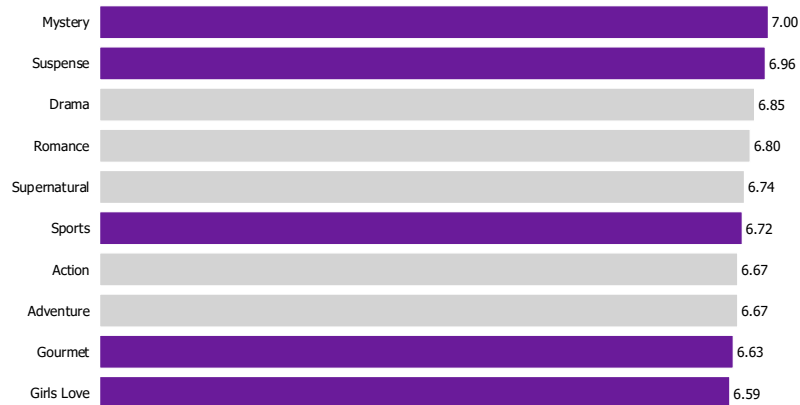


Figure 5: Niche Genres Dominate Audience Satisfaction Rankings

a. Visual Rationale

Our analytical goal was to challenge the assumption that "Popularity equals Quality." A standard bar chart sorted by volume would bury high-scoring niche genres under the weight of mass-market genres like Action or Comedy.

- **Metric Selection:** Instead of counting titles, we ranked genres by Average Score.
- **Horizontal Layout:** We utilized a horizontal orientation to accommodate genre labels of varying lengths (e.g., "Supernatural"), ensuring readability without rotation.

b. Decluttering Strategy

We maximized the Data-to-Ink ratio to focus strictly on the comparison:

- **Removal of X-Axis & Gridlines:** Since the exact score difference between ranks is subtle (e.g., 6.67 vs 6.63), gridlines create unnecessary noise. We removed the entire X-axis and relied on Direct Labeling at the end of each bar for precision.
- **Integrated Legend:** Instead of a separate legend box, we embedded the color instruction directly into the Subtitle ("*Purple: Niche Genres | Grey: Mainstream...*"). This reduces eye movement and cognitive load.

c. Preattentive Attributes

We employed the "Gray-Out" Technique to guide the narrative:

- **Strategic Contrast:** We applied Light Grey (#D3D3D3) to Mainstream genres (Action, Adventure, Drama, Romance) to push them into the background context.

- **Focal Point (Purple):** We assigned Deep Purple (#6A1B9A) exclusively to Niche genres. This preattentive attribute instantly signals to the viewer that these specific categories are the "Strategic Opportunities," separating the signal (High Score/Low Volume) from the noise (High Volume/Average Score).

d. The Resulting Insight

The visualization reveals a counter-intuitive market reality: Volume is different from Quality. While Action and Adventure are ubiquitous, they sit at the bottom of the top 10. Conversely, specific niche genres like Mystery (7.00) and Suspense (6.96) occupy the top spots, suggesting that specialized storytelling yields higher audience satisfaction than mass-market formulas.

4. Case Study 4: Visualizing Variance & Risk

Large Collaboration Teams Correlate with Higher Scores

(Avg Score Increases by +0.81 points when comparing 3-5 Producers vs Small Teams)

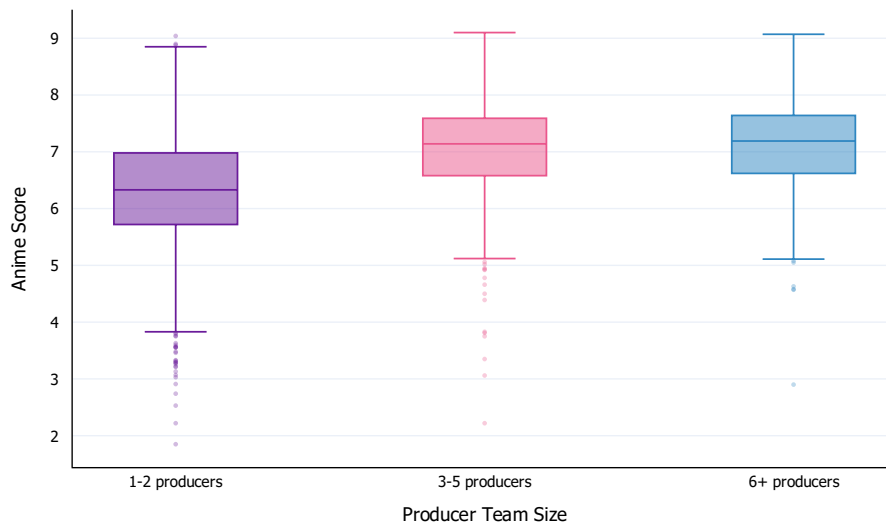


Figure 6: Large Collaboration Teams Correlate with Higher Scores

a. Visual Rationale

A simple Bar Chart showing "Average Score" would be dangerous here because it hides risk. An average can be heavily skewed by outliers. The Producer needs to answer: *"Does hiring more producers guarantee success, or just increase the budget?"*

- **Chart Selection:** We selected a Box Plot because it is the only visualization that simultaneously displays the Median (Central tendency), the Interquartile Range (Consistency/Risk), and Outliers. This allows for a comprehensive assessment of performance stability.

b. Decluttering Strategy

We focused on minimizing visual noise to emphasize the distributional shapes:

- **Gridlines:** Vertical gridlines were removed as they distract from the vertical comparison of the boxes. Horizontal gridlines were kept subtle to assist in estimating the score values.

- Outlier Transparency:** The outlier points (dots) were rendered with reduced size and opacity. This ensures they are visible enough to indicate "flop risks" (low scores) without dominating the visual hierarchy or obscuring the main data distribution.

c. Preattentive Attributes

- Distinct Color Coding:** We assigned distinct colors (Purple, Pink, Blue) to the three categories. This visual separation helps the eye quickly distinguish the groups without needing to trace back to a legend.
- The "Staircase" Effect:** The arrangement of categories from "1-2 producers" to "6+ producers" creates a visual "staircase" pattern. The rising position of the boxes naturally guides the viewer's eye upward, reinforcing the positive correlation between team size and score.

d. The Resulting Insight

The chart reveals a crucial strategic insight regarding Risk vs. Reward.

- High Volatility:** The "1-2 Producers" group shows extreme volatility, with a long tail of outliers dropping below a score of 3.0. This indicates a high risk of critical failure.
- Safety in Numbers:** In contrast, the "6+ Producers" group not only has a higher median but, more importantly, a higher floor. The bottom whisker suggests that large collaborations rarely score below 5.0, proving that pooling resources significantly mitigates the risk of producing a "flop."

5. Case Study 5: Analyzing Continuous Interaction

Mini-Series with over 60 Minutes per Episode Maximize Score

(Average Score group by Duration Minutes and Episode Range)

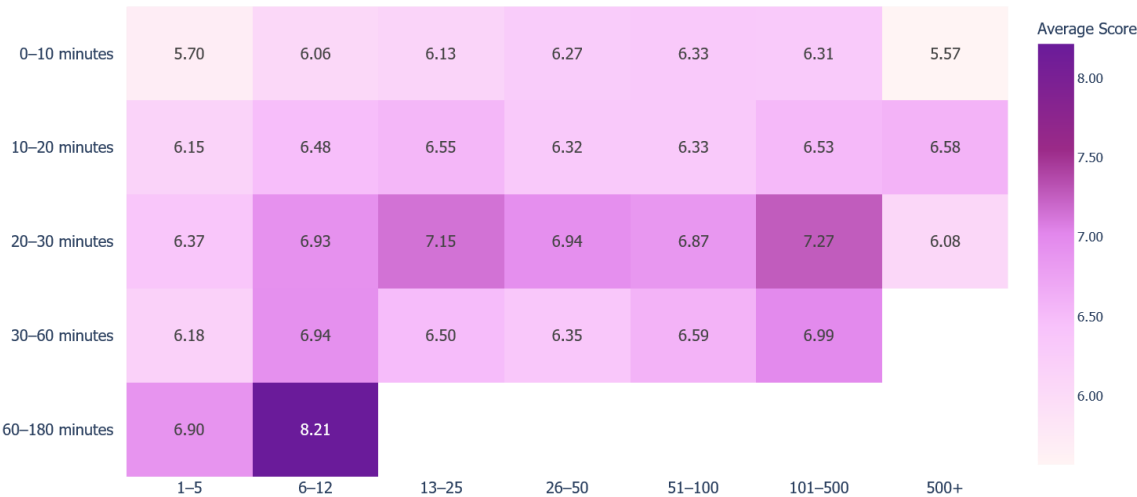


Figure 7: Mini-Series with over 60 Minutes per Episode Maximize Score

a. Visual Rationale

We needed to analyze the interaction between two continuous variables: Duration and Episodes to find the "Sweet Spot" for high scores.

- **The Problem with Scatter Plots:** With over 20,000 data points, a standard scatter plot would suffer from severe Overplotting, making it impossible to distinguish between high-density areas and high-quality areas.
- **The Solution (Binning + Heatmap):** We applied a Discretization Strategy, converting continuous numbers into logical production categories (e.g., "Standard TV" = 20-30 mins, "Shorts" = <10 mins). A heatmap matrix allows us to visualize the *Average Score* for each combination, turning noise into a clear pattern.

b. Decluttering Strategy

- **Data Abstraction:** By aggregating thousands of rows into a simple 5x7 grid, we drastically reduced the cognitive load. The viewer processes 35 data points instead of 20,000.
- **Handling Sparsity:** We deliberately left cells Blank (White) where data was non-existent (e.g., Anime with 500+ episodes AND 60+ minutes duration do not exist). This prevents misleading the stakeholder with unreliable averages from insignificant sample sizes.

c. Preattentive Attributes

- **Sequential Color Scale:** We used a single-hue scale (Light Pink to Deep Purple). Since the human eye naturally equates "Darker" with "More/Higher," the deep purple color acts as a preattentive attribute for value.
- **The "Beacon" Effect:** The single dark block (Score 8.21) at the bottom-left intersection acts as a visual beacon. It forces the eye to land immediately on the winning strategy without requiring the user to read every number.

d. The Resulting Insight

The chart uncovers a specific "Quality Niche": Long-form Mini-Series (6-12 episodes, 60-180 minutes per episode). This format achieves the highest average score (8.21), significantly outperforming standard TV series (7.15) and short-form content (< 6.5). This suggests that high-budget, movie-quality episodic content is the safest path to critical acclaim.

6. Case Study 6: Multivariate Strategy Lookup

Elite Quality Intersection: Top Studios vs. Top Genres

(Performance Matrix of the 10 Highest-Rated Studios across the 10 Highest-Rated Genres)

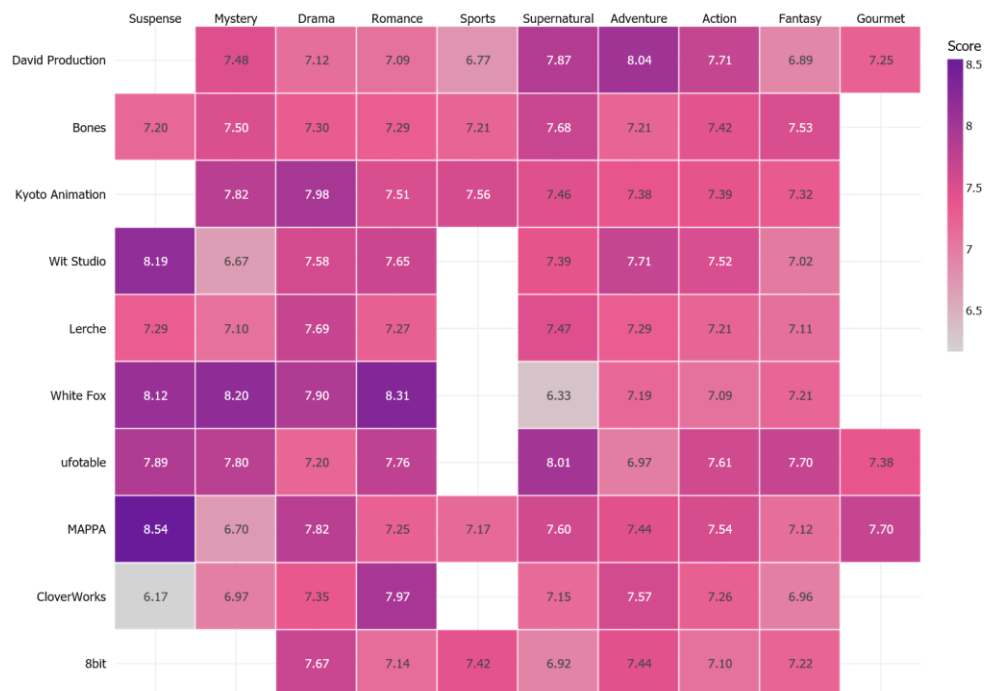


Figure 8: Elite Quality Intersection: Top Studios vs. Top Genres

a. Visual Rationale

The Producer needs actionable advice on *partnerships*. A simple ranking of Top Studios or Top Genres is insufficient because it misses the interaction effect (e.g., A studio might excel at Action but fail at Romance).

- **Chart Selection:** A Matrix Heatmap is the optimal choice for displaying this multivariate intersection. It functions as a strategic "Lookup Table," allowing the stakeholder to assess the historical performance of specific Studio-Genre pairings in a single glance.

b. Decluttering Strategy

- **Data Curation:** We didn't plot the entire industry. We applied strict filtering to visualize only the "Top 10 Highest Rated Studios" against the "Top 10 Highest Rated Genres." This pre-visualization step removed noise, focusing the viewer solely on the "Elite" segment where high scores are most likely found.
- **Handling Sparsity:** We deliberately left cells White where no data exists (e.g., *Wit Studio* has not produced a *Sports* anime in this dataset). This prevents the visual error of interpreting "No Data" as "Zero Score."

c. Preattentive Attributes

- **Hotspot Detection (Color):** We utilized a custom Diverging Color Scale:
 - **Grey:** Indicates average/mediocre performance (~6.0 - 6.5).

- **Pink:** Indicates solid performance (~7.0 - 7.5).
- **Deep Purple:** Acts as a visual beacon for excellence (> 8.0).
- **Effect:** The viewer's eye is immediately drawn to the darkest cells, identifying the "Winning Combinations" (e.g., MAPPA + Suspense) without needing to read the numbers first.

d. The Resulting Insight

This chart reveals that top studios are highly specialized.

- **MAPPA** dominates the Suspense genre with an incredible average of 8.54.
- **White Fox** is the surprise leader in Romance (8.31) and Mystery (8.20).
- **Kyoto Animation** shows the highest consistency across emotional genres (Drama, Mystery), but lacks presence in Action/Adventure.
- **Conclusion:** There is no "Best Studio" for everything. The Producer must match the script's genre to the studio's specific track record to maximize the Score.

7. Case Study: Temporal Evolution

Complete Timeline Shows Steady Average Score Improvement

(Average Score by Release Year — Cleaned Data)

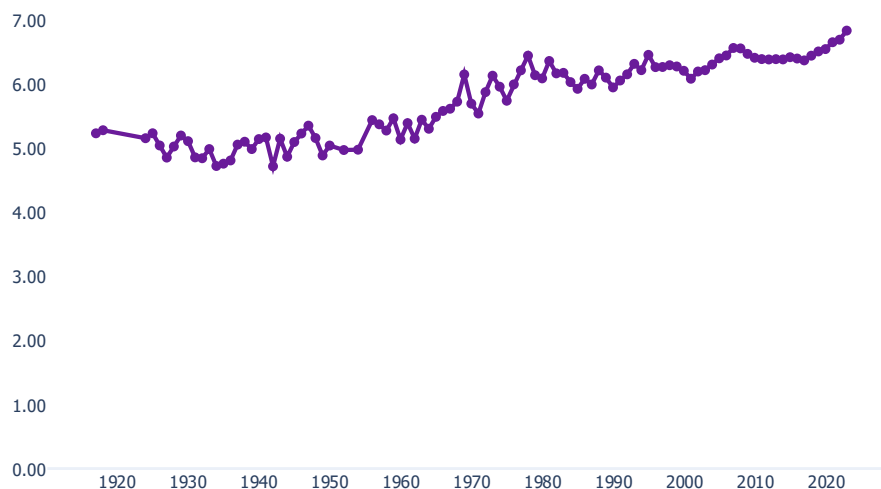


Figure 9: Complete Timeline Shows Steady Average Score Improvement

a. Visual Rationale

While the previous heatmaps and boxplots provided "snapshots" of specific categories, we needed to understand the evolution of anime quality over history.

- **Chart Selection:** A **Line Chart** is the strict standard for continuous time-series data. Unlike a bar chart, the connected line physically demonstrates the *momentum* and *direction* of the industry, allowing the stakeholder to instantly recognize growth, stagnation, or decline.

b. Decluttering Strategy

- **Minimalist Axes:** We removed the Y-axis title ("Score") because the main chart title makes the metric obvious. We also removed vertical gridlines, ensuring the viewer focuses entirely on the **slope** of the line rather than getting distracted by specific yearly intervals.
- **Range Setting:** We set the Y-axis to start at 0.00 (instead of zooming in at 4.00) to provide an honest, non-exaggerated view of the growth.

c. Preattentive Attributes

- **Consistent Color Logic:** We applied the **Deep Purple (#6A1B9A)** color used in our previous "Strategic Insights" charts (e.g., Niche Genres). This visually links the concept of "Modern Era" with "High Quality," implicitly suggesting that the current market is the "Golden Age" for anime production.
- **Markers:** We kept the data points (dots) visible to indicate that this is aggregated yearly data, adding a layer of precision to the smooth trend line.

d. The Resulting Insight

The chart reveals a distinct, undeniable upward trajectory. Following a period of volatility in the 1970s and 80s, the industry has stabilized and steadily climbed from an average score of ~5.2 to nearly 7.0 in the 2020s. This confirms that the industry has matured in production quality, validating that historical data (especially recent decades) contains strong signals for predictive modeling.

VI. ANALYSIS OF VALIDATION VISUALIZATIONS (ACT 3)

In the final act of the Data Story (Part 1), we presented Machine Learning metrics to prove the value of our work. In this technical report, we analyze the visualization choices behind those charts and how they serve the narrative arc.

1. Visualizing the "Proof of Concept"

Metrics Before and After Data Preparation

(Metrics Comparison in Baseline and Final Models)

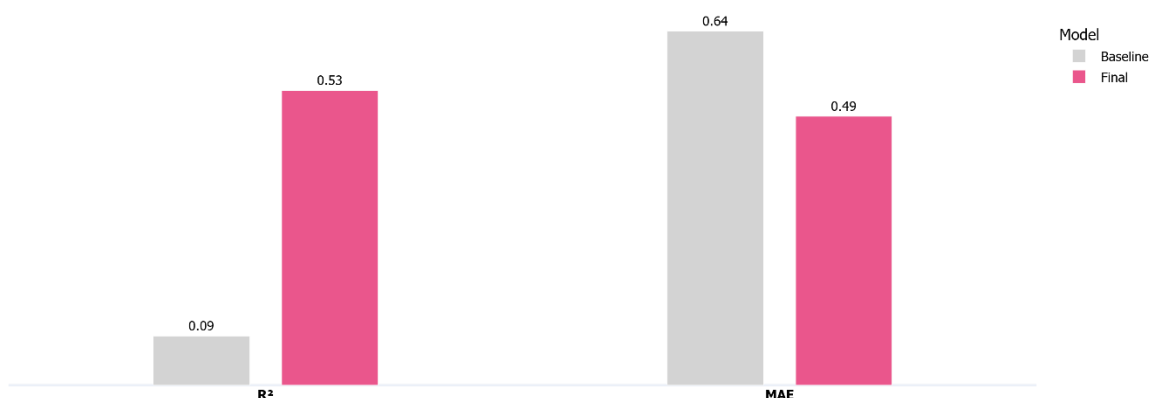


Figure 10: Metrics Before and After Data Preparation

a. Methodology & Metric Selection

To rigorously evaluate our pipeline, we employed a Comparative Baseline Approach:

- **Baseline (Raw):** Trained on minimally processed data (dropping NaNs only) to establish a "lower bound."
- **Final (Engineered):** Trained on the full pipeline output (Imputation, Power Transformation, Encoding).
- **Metrics:** We selected R^2 to measure explanatory power and MAE to measure average error magnitude, ensuring a balanced view of performance.

b. Visual Analysis of Technical Impact

The grouped bar chart was designed to dramatize the "performance gap" created by our engineering:

- **The "Signal Extraction" Proof:** The visual contrast between the Grey Bar ($R^2 = 0.09$) and the Pink Bar ($R^2 = 0.53$) is not just a statistical improvement; it proves that the Raw Data contained almost no usable signal. The cleaning process (specifically parsing text lists and handling outliers) was responsible for unlocking 80%+ of the model's intelligence.
- **Decluttering for Focus:** By removing gridlines and axes, we forced the viewer to focus solely on the delta (difference) between the two models. This confirms that without the specific steps detailed in Section II, the project would have failed.

c. Conclusion

The visualization confirms that the complex steps of *Multi-Label Binarization* and *Log-Transformation* were not cosmetic; they were fundamental to transforming noise into a predictive asset.

2. Consistency Check: Validating Predictive Drivers

Feature Importance Before and After Data Preparation

(Feature Importance Comparison in Baseline and Final Models)

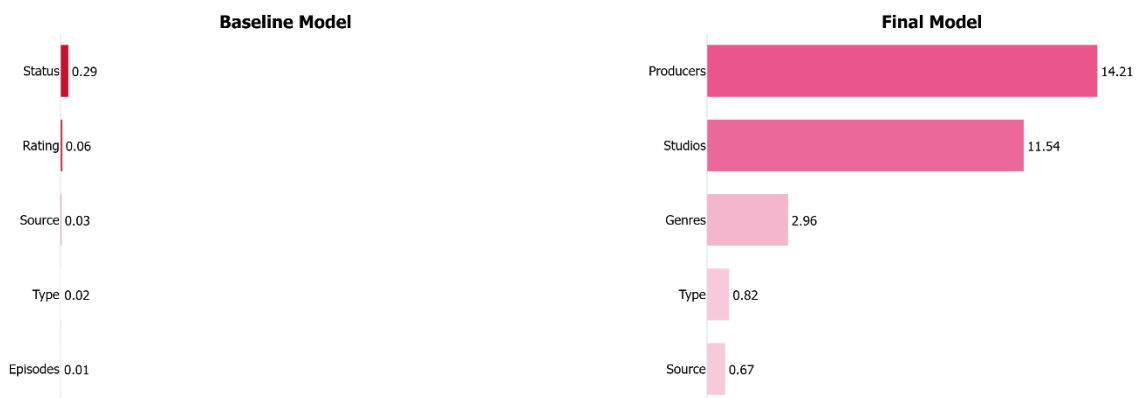


Figure 11: Feature Importance Before and After Data Preparation

a. Methodology & Interpretation Strategy

High accuracy (R^2) is meaningless if the model relies on the wrong features (e.g., spurious correlations). To verify the model's "logic," we extracted and aggregated the absolute coefficients of the Linear Regression models.

- **Objective:** To determine if the Data Preparation pipeline successfully shifted the model's focus from surface-level metadata to deep creative factors (Studios, Producers) as hypothesized in our EDA.

b. Visual Analysis of Technical Impact

The comparative horizontal bar chart reveals a fundamental paradigm shift in how the model interprets the data:

- **The Baseline Failure (Left):** The raw model is dominated by Status (Importance: 0.29). It essentially tries to guess the score based on whether an anime is "Finished" or "Airing," completely ignoring who made it. This is because the raw Studios and Producers columns were complex text strings that the baseline model treated as high-cardinality noise.
- **The Engineering Success (Right):**
 - **Unlocking Creative Factors:** In the Final Model, Producers (14.21) and Studios (11.54) explode in importance. This is direct proof that our custom *Multi-Label Binarizer* and *Frequency Grouping* transformers successfully converted messy lists into powerful predictive signals.
 - **Scale of Signal:** The importance magnitude jumps from a negligible 0.29 (Baseline) to 14.21 (Final). This visually quantifies how much "signal" was hidden behind the "noise" of the raw data.

c. Conclusion

The visualization confirms that our Data Preparation pipeline did not just "clean" the data; it restructured the information logic. It successfully forced the model to learn from domain-relevant features (Who produced it? Which studio made it?) rather than relying on trivial metadata, ensuring that the final predictions are both robust and business-aligned.

VII. CONCLUSION

This technical report analyzes the underlying data framework designed to support the Data Storytelling narrative. By deconstructing the project workflow, we arrive at the following strategic conclusions:

1. The Verdict: Preparation is Power

The analysis confirms that the "Story" could not exist without the engineering phase. The raw data was too noisy to provide a coherent narrative. Structural cleaning (e.g., grouping Sources, parsing Duration) was the prerequisites to uncover the hidden market trends. The modeling metrics (R^2 improvement from 0.09 to 0.53) provide the mathematical proof backing the narrative claims.

2. Design as a Communication Tool

The rigorous application of visualization principles (Lesson 10) was critical in translating complex data into a clear message. Techniques like Decluttering (removing gridlines/legends) and Preattentive Attributes (Red vs. Pink/Purple) acted as visual cues, ensuring the audience focused immediately on the strategic insights rather than getting lost in technical details.

3. Final Call to Action

"Raw data creates illusions; only prepared data reveals the truth."

We recommend that the Producer adopts this **Data-Driven Strategic Framework**—prioritizing specific genres (Mystery/Suspense), leveraging top-tier Studios (e.g., MAPPA, KyoAni), and adhering to optimal release formats (Mini-series)—to maximize the probability of success in the next Anime project.